

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]
```

```

cols = list(dict.fromkeys(target_col + features_X +
                          feats_to_balance + feats_grouping +
                          feats_reference+ feats_postproc))

df = df[cols]
#df['campaign'] = "ALL"

return df

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r") as f:
            #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r") as f:
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")

    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)

```

```
#Removing of rabbit 4
df["column"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)

return df,dict_features
```

```
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Function print :

1.2 Metadata

Nombre de lignes df

8032

Nombre de lignes df clean	8032
Nombre de features	26
% de NaN	0.0
Taille dataset entraînement	5622
Taille dataset test	2409
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_3h_D1_energy, r_NH3_wl_3h_D2_energy, r_NH3_wl_3h_A_energy, r_NH3_wl_3h_A_rms, r_NH3_wl_3h_D1_energy_rel, r_NH3_wl_3h_D2_energy_rel, r_NH3_wl_3h_A_energy_rel, r_temperature_wl_3h_D1_energy, r_temperature_wl_3h_D2_energy, r_temperature_wl_3h_A_energy, r_temperature_wl_3h_A_rms, r_temperature_wl_3h_D1_energy_rel, r_temperature_wl_3h_D2_energy_rel, r_temperature_wl_3h_A_energy_rel, r_umidity_wl_3h_D1_energy, r_umidity_wl_3h_D2_energy, r_umidity_wl_3h_A_energy, r_umidity_wl_3h_A_rms, r_umidity_wl_3h_D1_energy_rel, r_umidity_wl_3h_D2_energy_rel, r_umidity_wl_3h_A_energy_rel

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermap and cluster list

The following section has been done with a threshold choosen of : 0.8

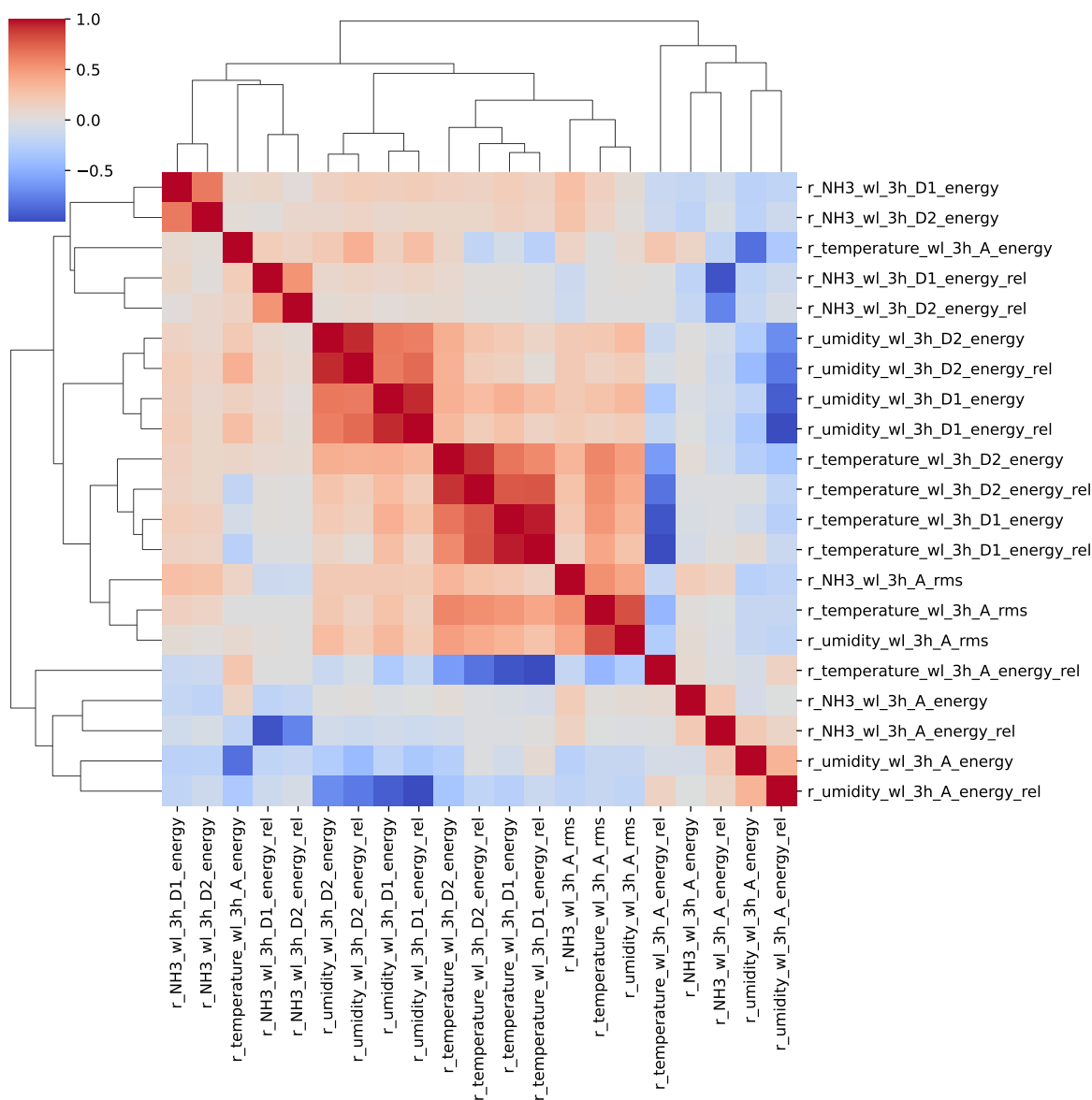
cluster 1: r, , N, H, 3 ; **cluster 2:** r, , t, e, m, p, e, r, a, t, u, r, e ; **cluster 3:** r, , u, m, i, d, i, t, y ; **cluster 4:** r, , C, O, 2 ; **cluster 5:** r, _, T, H, I ; **cluster 6:** r_NH3_wl_3h_D1_energy ; **cluster 7:** r_NH3_wl_3h_D2_energy ; **cluster 8:** r_NH3_wl_3h_A_energy ; **cluster 9:** r_NH3_wl_3h_A_rms ; **cluster 10:** r_NH3_wl_3h_D1_energy_rel, r_NH3_wl_3h_A_energy_rel ; **cluster 11:** r_NH3_wl_3h_D2_energy_rel ; **cluster 12:** r_temperature_wl_3h_D2_energy_rel, r_temperature_wl_3h_D2_energy, r_temperature_wl_3h_D1_energy_rel, r_temperature_wl_3h_A_energy_rel, r_temperature_wl_3h_D1_energy ; **cluster 13:**

r_temperature_wl_3h_A_energy, r_umidity_wl_3h_A_energy ; **cluster 14:** r_temperature_wl_3h_A_rms, r_umidity_wl_3h_A_rms ; **cluster 15:** r_umidity_wl_3h_A_energy_rel, r_umidity_wl_3h_D1_energy_rel, r_umidity_wl_3h_D1_energy ; **cluster 16:** r_umidity_wl_3h_D2_energy, r_umidity_wl_3h_D2_energy_rel ;

2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Chooosen features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_3h_D1_energy, r_NH3_wl_3h_D2_energy, r_NH3_wl_3h_A_energy, r_NH3_wl_3h_A_rms, r_NH3_wl_3h_D1_energy_rel, r_NH3_wl_3h_D2_energy_rel, r_temperature_wl_3h_D2_energy_rel, r_temperature_wl_3h_A_energy, r_temperature_wl_3h_A_rms, r_umidity_wl_3h_A_energy_rel, r_umidity_wl_3h_D2_energy



3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_3h_D1_energy, r_NH3_wl_3h_D2_energy, r_NH3_wl_3h_A_energy, r_NH3_wl_3h_A_rms,

r_NH3_wl_3h_D1_energy_rel, r_NH3_wl_3h_D2_energy_rel, r_temperature_wl_3h_D2_energy_rel, r_temperature_wl_3h_A_energy, r_temperature_wl_3h_A_rms, r_umidity_wl_3h_A_energy_rel, r_umidity_wl_3h_D2_energy

Len(Features_X): 16

Config Gridsearch:{‘show’: True, ‘search_type’: ‘optuna’, ‘model_name’: None, ‘model’: None, ‘param_grid’: None, ‘best_param’: None, ‘cv_func’: ShuffleSplit(n_splits=5, random_state=42, test_size=0.3, train_size=None), ‘save_models’: True, ‘sumup-table’: True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_{\text{test}} < 0.95$ (under performance of the test)

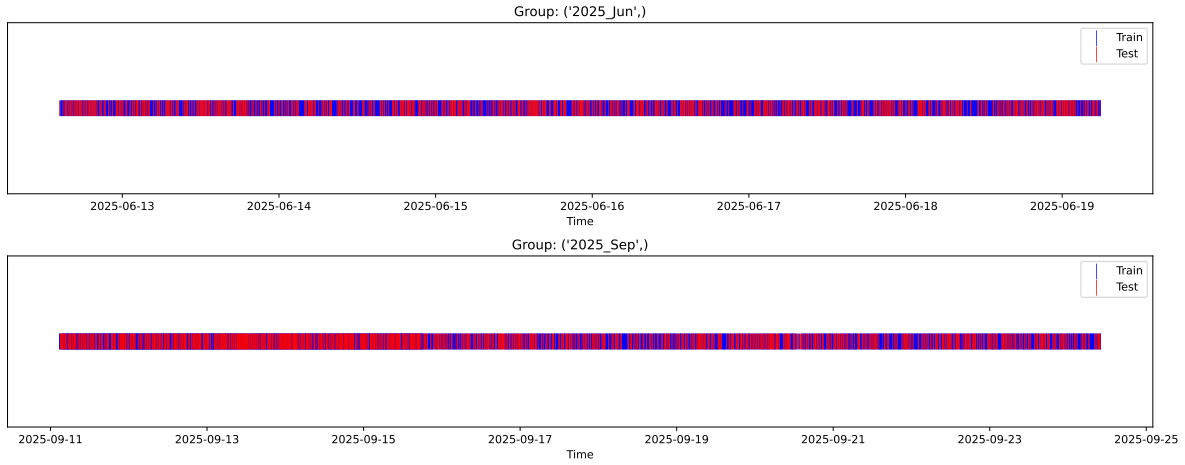
[Green] $r_{\text{test}} > 1.05$ (over performance of the test)

[Red] r_{train} not in $[0.95, 1.05]$ (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_train vs X_test

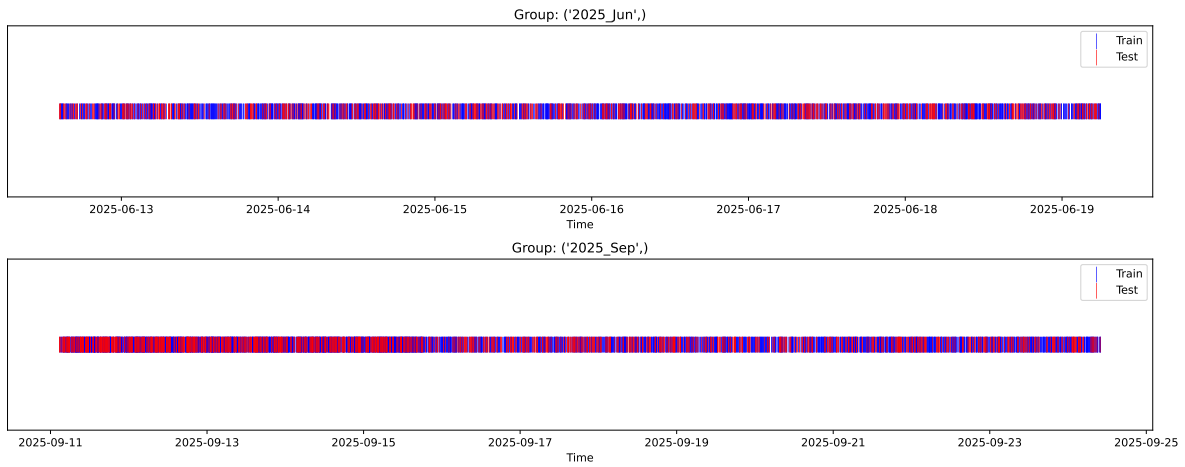
External Split: Train: 5622 samples Test: 2410 samples



3.3.2 Internal CV Folds

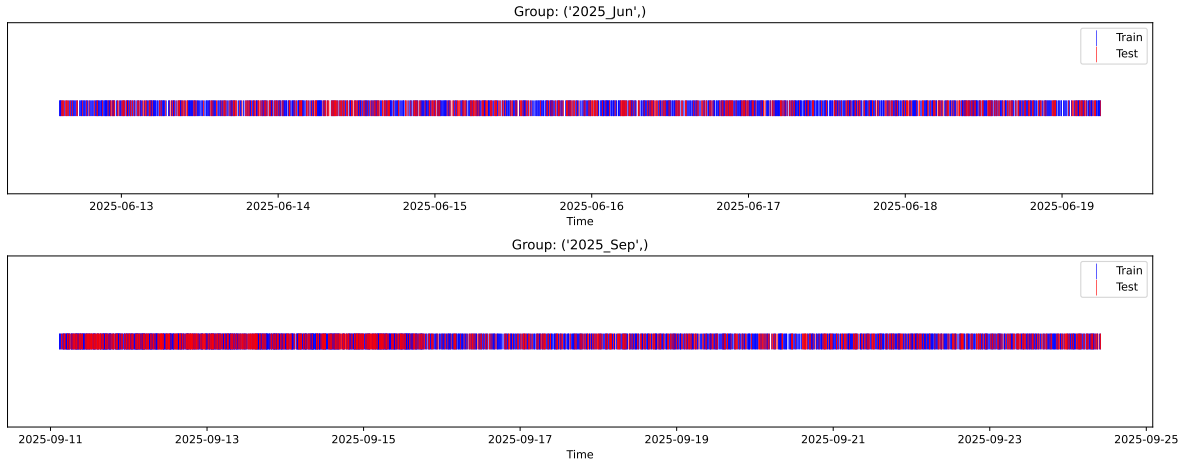
Fold 1/5

Train: 3935 samples Test: 1687 samples Excluded: 0 samples Window exclusion: 0h



Fold 2/5

Train: 3935 samples Test: 1687 samples Excluded: 0 samples Window exclusion: 0h



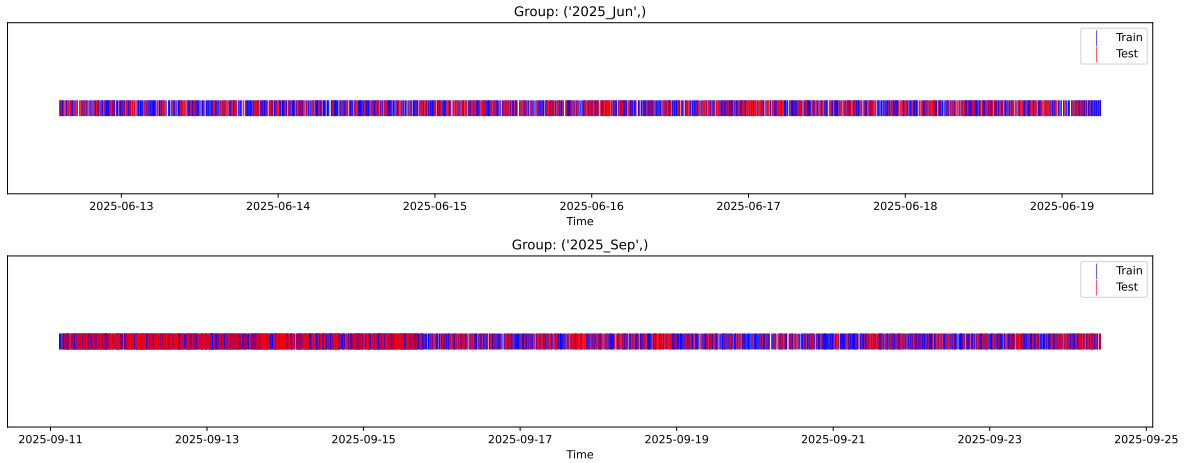
Fold 3/5

Train: 3935 samples Test: 1687 samples Excluded: 0 samples Window exclusion: 0h



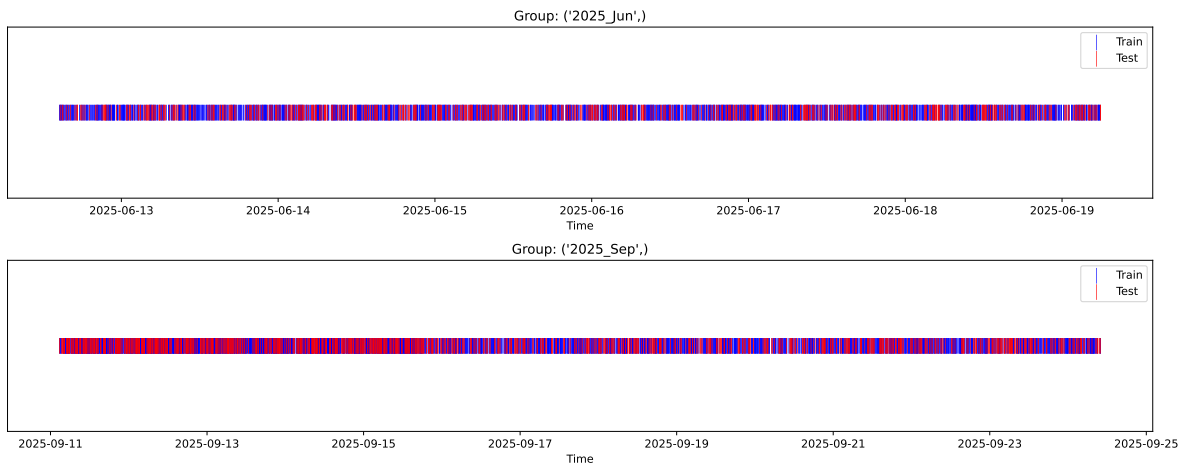
Fold 4/5

Train: 3935 samples Test: 1687 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3935 samples Test: 1687 samples Excluded: 0 samples Window exclusion: 0h



3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.346, std=0.800, min=0.959, max=13.845
 y_{test} : mean=2.357, std=0.789, min=1.070, max=8.190

Fold 0 — y_{test} : mean=2.34, std=0.80, y_{train} : mean=2.35, std=0.80

Fold 1 — y_{test} : mean=2.32, std=0.76, y_{train} : mean=2.36, std=0.82

Fold 2 — y_{test} : mean=2.35, std=0.79, y_{train} : mean=2.34, std=0.80

Fold 3 — y_test: mean=2.35, std=0.81, y_train: mean=2.35, std=0.80

Fold 4 — y_test: mean=2.33, std=0.83, y_train: mean=2.35, std=0.78

train scores CV: [0.52937431 0.53532033 0.51984676 0.556931 0.55493624]

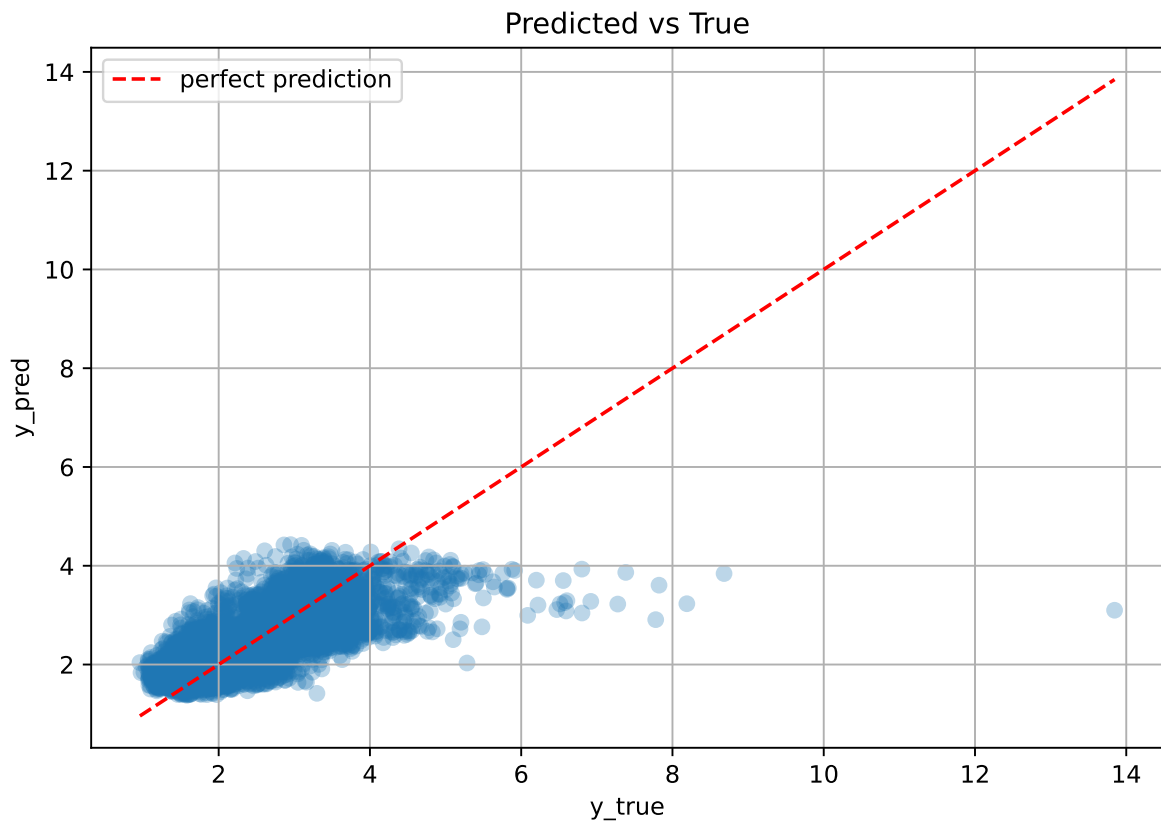
test scores CV: [0.55153961 0.53519441 0.57468249 0.48677642 0.49677822]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.5393	0.529	0.033	0.561	0.5615	0.0325	1.0614	1.0194

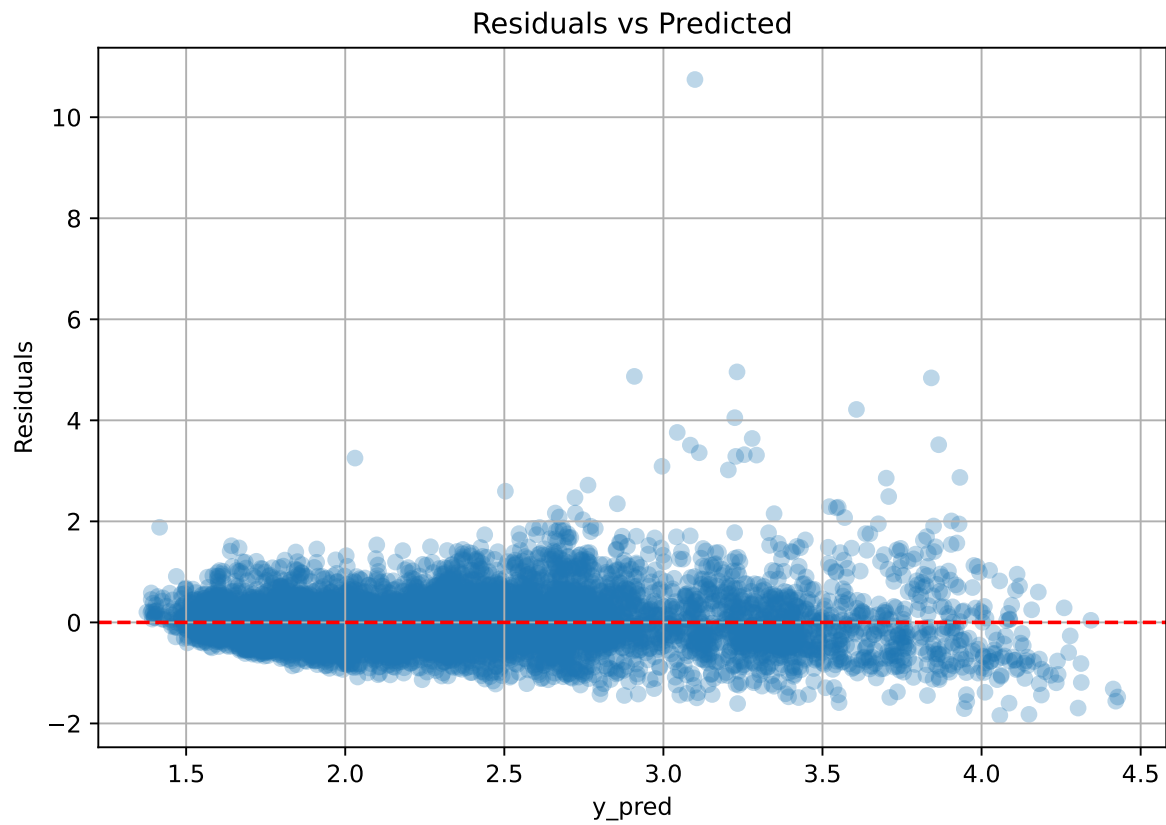
Coefficients:

coef_r_CO2 = -0.0447, coef_r_NH3 = 0.2256, coef_r_NH3_wl_3h_A_energy = -0.2930, coef_r_NH3_wl_3h_A_rms = 0.0087, coef_r_NH3_wl_3h_D1_energy = 0.0402, coef_r_NH3_wl_3h_D1_energy_rel = -0.0474, coef_r_NH3_wl_3h_D2_energy = -0.0105, coef_r_NH3_wl_3h_D2_energy_rel = -0.0369, coef_r_THI = 0.5557, coef_r_temperature = -0.0727, coef_r_temperature_wl_3h_A_energy = -0.5629, coef_r_temperature_wl_3h_A_rms = 0.1032, coef_r_temperature_wl_3h_D2_energy_rel = -0.0316, coef_r_umidity = -0.5106, coef_r_umidity_wl_3h_A_energy_rel = -0.0080, coef_r_umidity_wl_3h_D2_energy = -0.0109, intercept = 2.3460,

3.4.2 Scatter



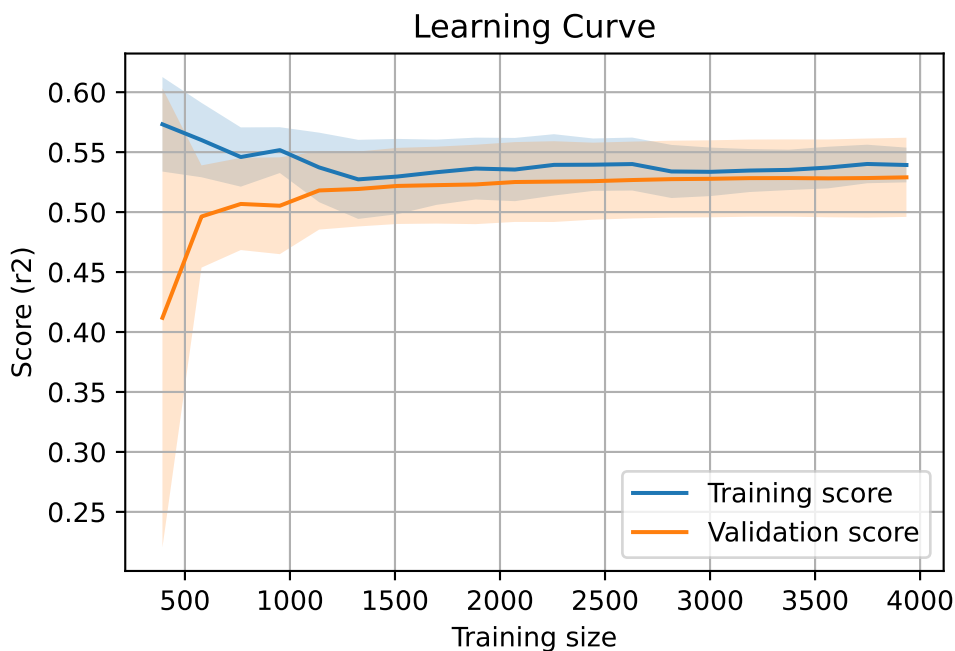
3.4.3 Scatter residuals



3.4.4 Learning curve

`Len(group)` : 8032, `Len(training)` : 5623,

`Len(training_real)` : 3937, `Len(test_of_training)` : 1686, `Len(test)` : 2409



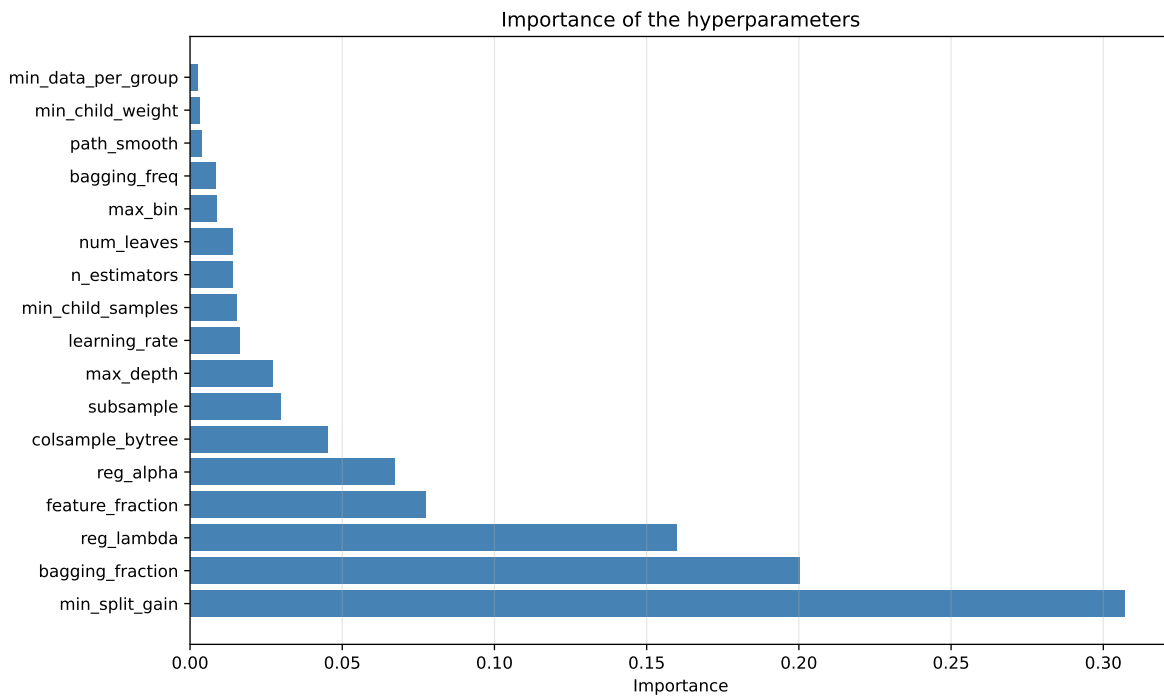
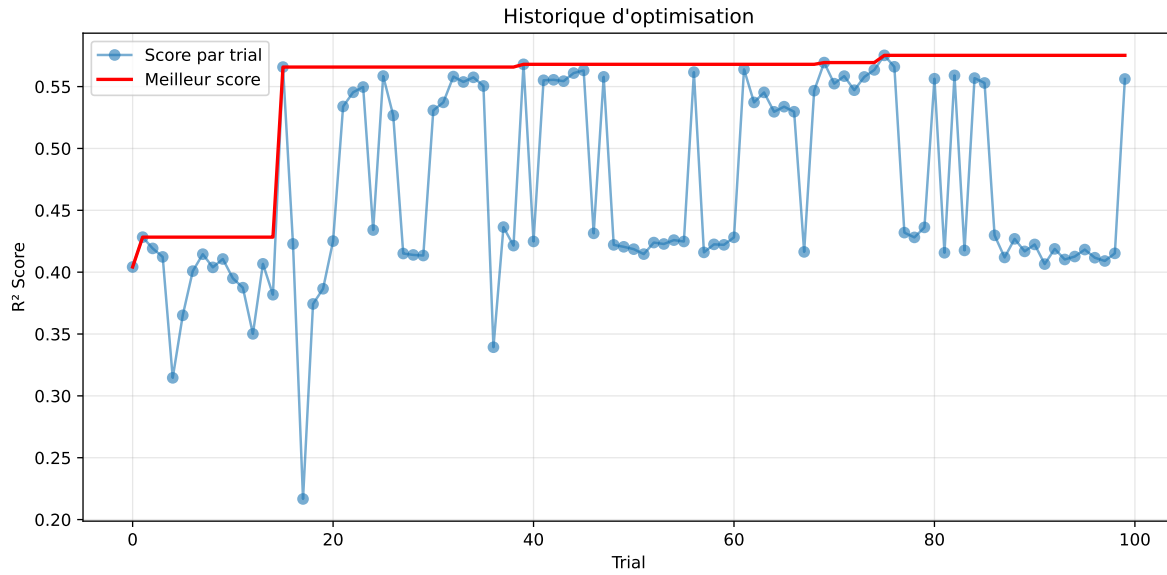
3.5 Model : LGBM

3.5.1 Gridsearch

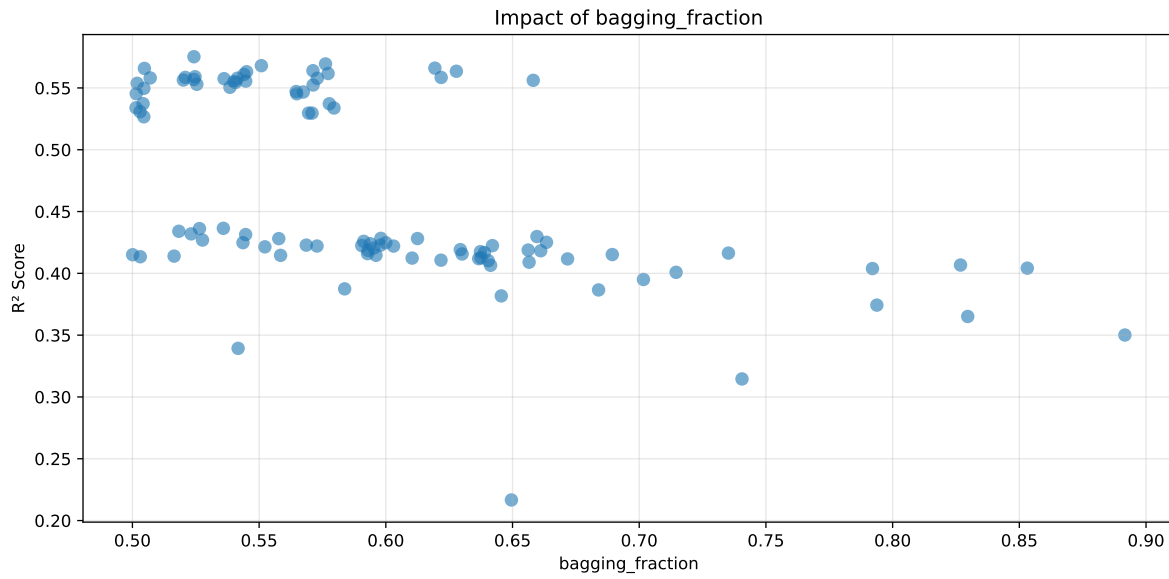
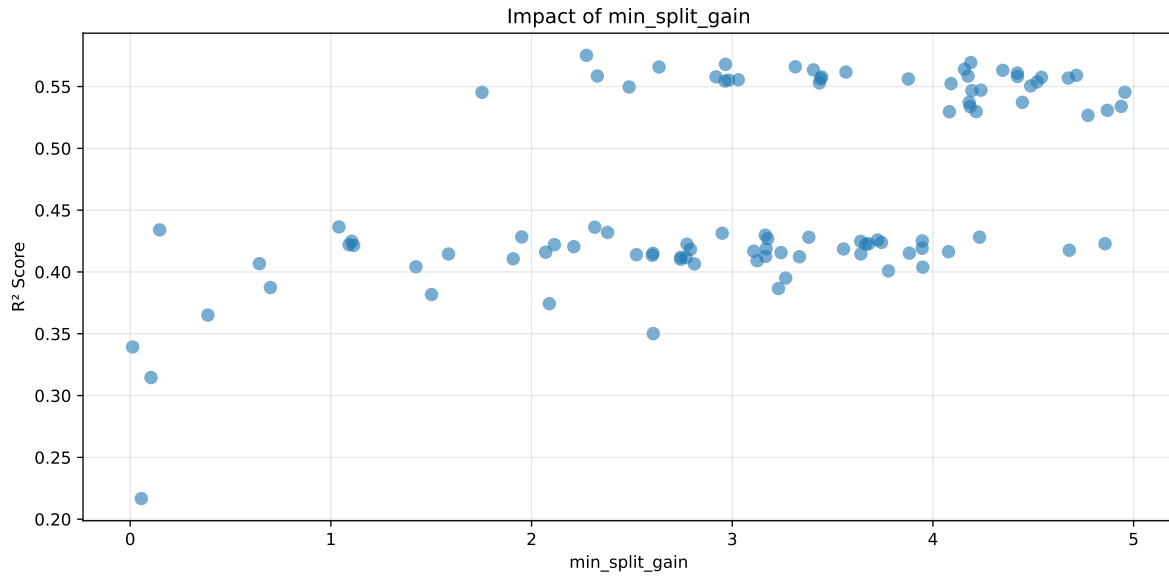
Distribution y_train vs y_test: y_train: mean=2.346, std=0.800, min=0.959, max=13.845
y_test: mean=2.357, std=0.789, min=1.070, max=8.190

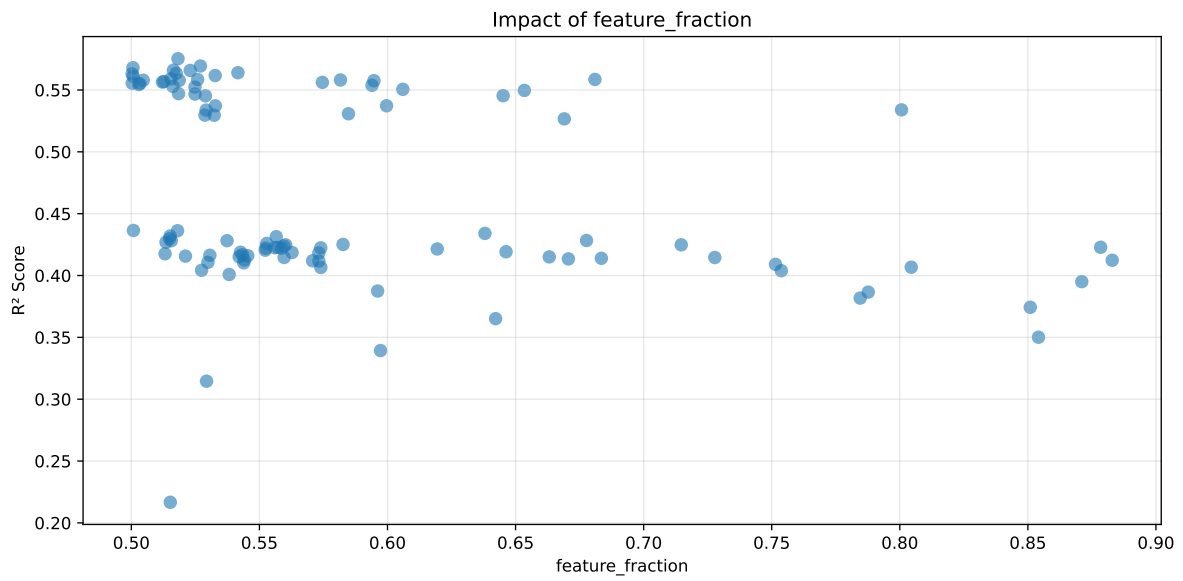
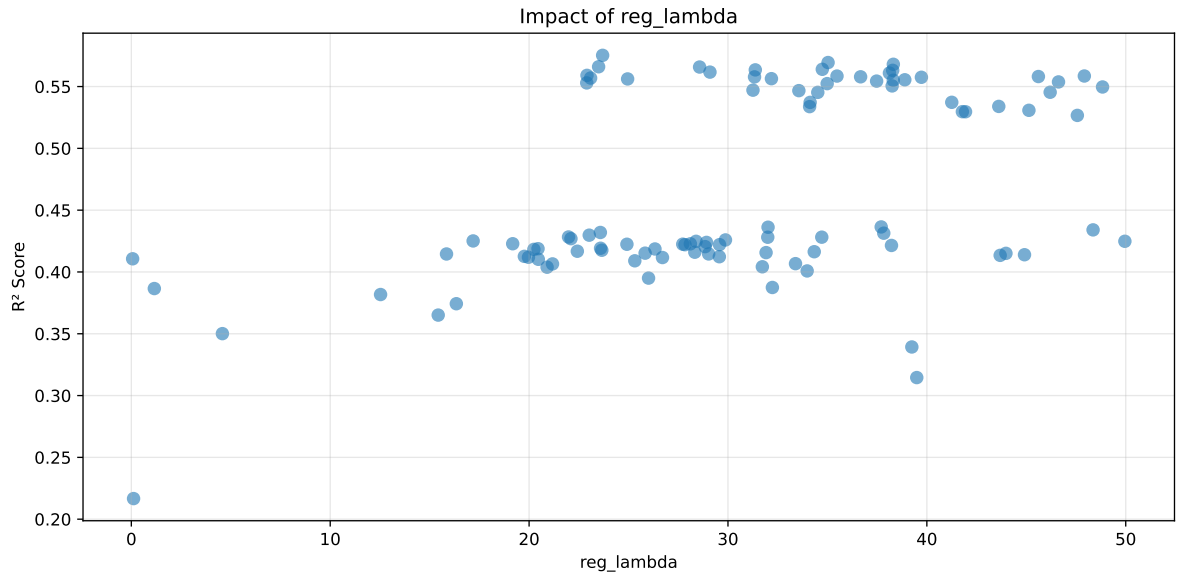
===== TOP Importance FEATURES =====

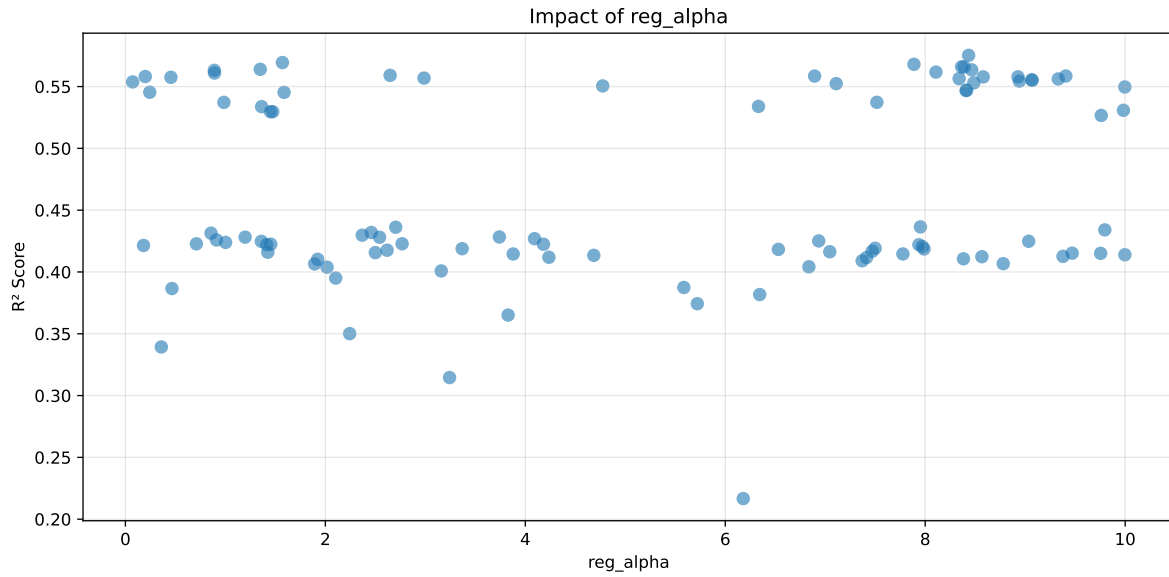
feature importance	feature importance	feature importance	feature importance	feature importance
r_CO2	156	r_umidity	124	r_temperature_wl_3h_A_rms
122	r_temperature_wl_3h_A_energy	103	r_temperature	96
r_temperature_wl_3h_D2_energy_rel	75	r_NH3_wl_3h_D1_energy	68	r_NH3_wl_3h_D2_energy
67	r_NH3	59	r_THI	56
r_NH3_wl_3h_A_energy	56	r_NH3_wl_3h_D2_energy_rel	41	r_NH3_wl_3h_A_rms
39	r_umidity_wl_3h_A_energy_rel	34	r_umidity_wl_3h_D2_energy	18
r_NH3_wl_3h_D1_energy_rel	12			



Paramètres avec >5% d'importance : ['min_split_gain', 'bagging_fraction', 'reg_lambda', 'feature_fraction', 'reg_alpha']

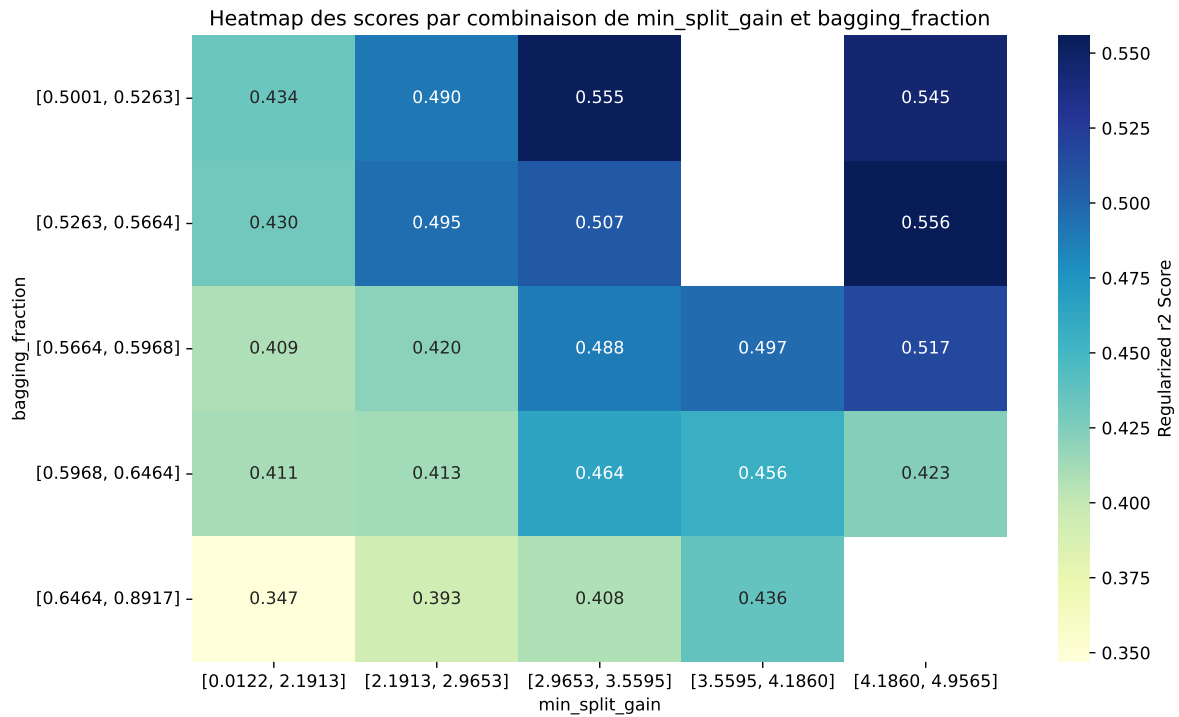






C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6024	0.5753	0.5958	0.0307	0.5753	0.6202	0.0449	1.078	1.0471
2	0.5973	0.5694	0.5886	0.0319	0.5694	0.6099	0.0405	1.0711	1.049
3	0.5938	0.568	0.5866	0.0316	0.568	0.6087	0.0407	1.0716	1.0454
4	0.5934	0.566	0.5855	0.029	0.566	0.6083	0.0423	1.0747	1.0484
5	0.5934	0.5658	0.5865	0.0318	0.5658	0.6255	0.0597	1.1055	1.0488
6	0.5919	0.564	0.5835	0.0337	0.564	0.6061	0.0421	1.0747	1.0496
7	0.5895	0.5635	0.5824	0.0314	0.5635	0.6034	0.0399	1.0708	1.0461
8	0.5893	0.5631	0.5823	0.0312	0.5631	0.6031	0.04	1.071	1.0465
9	0.5885	0.5617	0.5803	0.0319	0.5617	0.6049	0.0432	1.0769	1.0477
10	0.5868	0.5609	0.5803	0.0308	0.5609	0.6021	0.0412	1.0734	1.0461
---	---	---	---	---	---	---	---	---	---
100	0.7997	0.7025	0.7313	0.0324	0.2167	0.7463	0.0438	1.0623	1.1383

Hyperparameters:

Rank 1: {'reg__n_estimators': 300, 'reg__max_depth': 3, 'reg__learning_rate': 0.023042191999660892, 'reg__num_leaves': 60, 'reg__subsample': 0.7484479520065198, 'reg__colsample_bytree': 0.7239521902693847, 'reg__min_child_samples': 16, 'reg__reg_alpha': 8.434262770951255, 'reg__reg_lambda': 23.69755353287264, 'reg__min_split_gain': 2.273386696346752, 'reg__path_smooth': 3.3365578837079743, 'reg__min_child_weight': 0.07906002430048255, 'reg__feature_fraction': 0.5182188589015236, 'reg__bagging_fraction': 0.524271298141654, 'reg__bagging_freq': 3, 'reg__max_bin': 125, 'reg__min_data_per_group': 139}

Rank 2: {'reg__n_estimators': 200, 'reg__max_depth': 6, 'reg__learning_rate': 0.02171955995513848, 'reg__num_leaves': 23, 'reg__subsample': 0.7412730043540888, 'reg__colsample_bytree': 0.7516069828122806, 'reg__min_child_samples': 12, 'reg__reg_alpha': 1.5705464834417264, 'reg__reg_lambda': 35.03335681589824, 'reg__min_split_gain': 4.18876095252126, 'reg__path_smooth': 4.307556628904169, 'reg__min_child_weight': 0.5430859523839087, 'reg__feature_fraction': 0.526973021237291, 'reg__bagging_fraction': 0.5762016179786978, 'reg__bagging_freq': 7, 'reg__max_bin': 151, 'reg__min_data_per_group': 158}

Rank 3: {'reg__n_estimators': 500, 'reg__max_depth': 4, 'reg__learning_rate': 0.019185888260031883, 'reg__num_leaves': 49, 'reg__subsample': 0.671419371864067, 'reg__colsample_bytree': 0.6980745797843312, 'reg__min_child_samples': 15, 'reg__reg_alpha': 7.88727458294114, 'reg__reg_lambda': 38.31474614154646, 'reg__min_split_gain': 2.966301059513442, 'reg__path_smooth': 3.9315038621391967, 'reg__min_child_weight': 0.1529749333433689, 'reg__feature_fraction': 0.5006372702523307, 'reg__bagging_fraction': 0.5508880015658046, 'reg__bagging_freq': 5, 'reg__max_bin': 101, 'reg__min_data_per_group': 148}

Rank 4: {'reg__n_estimators': 300, 'reg__max_depth': 3, 'reg__learning_rate': 0.023886075642636152, 'reg__num_leaves': 60, 'reg__subsample': 0.7498118221697498, 'reg__colsample_bytree': 0.725033154057661, 'reg__min_child_samples': 17, 'reg__reg_alpha': 8.363994147658264, 'reg__reg_lambda': 23.49491087157741, 'reg__min_split_gain': 3.314336988228291, 'reg__path_smooth': 3.371259368438657, 'reg__min_child_weight': 0.07773489158140193, 'reg__feature_fraction': 0.5164440139544165, 'reg__bagging_fraction': 0.6193537332199535, 'reg__bagging_freq': 3, 'reg__max_bin': 153, 'reg__min_data_per_group': 137}

Rank 5: {'reg__n_estimators': 400, 'reg__max_depth': 7, 'reg__learning_rate': 0.02984016728681039, 'reg__num_leaves': 56, 'reg__subsample': 0.7418831674867169, 'reg__colsample_bytree': 0.5958434233926623, 'reg__min_child_samples': 11, 'reg__reg_alpha': 8.388981208459441, 'reg__reg_lambda': 28.57320035606771, 'reg__min_split_gain': 2.6353557923244724, 'reg__path_smooth': 3.4606872451625876, 'reg__min_child_weight': 0.08697728294695031, 'reg__feature_fraction': 0.522995349986367, 'reg__bagging_fraction': 0.5047271209425062, 'reg__bagging_freq': 4, 'reg__max_bin': 102, 'reg__min_data_per_group': 108}

Rank 6: {'reg__n_estimators': 200, 'reg__max_depth': 3, 'reg__learning_rate': 0.023191163796284033, 'reg__num_leaves': 24, 'reg__subsample': 0.6351498537443604, 'reg__colsample_bytree': 0.7599068545228345, 'reg__min_child_samples': 13, 'reg__reg_alpha': 1.348881480334209, 'reg__reg_lambda': 34.74303279286167, 'reg__min_split_gain': 4.157253106468791, 'reg__path_smooth': 4.213354707439603, 'reg__min_child_weight': 0.5482907619880019, 'reg__feature_fraction': 0.5416136659950637, 'reg__bagging_fraction': 0.5712464640822003, 'reg__bagging_freq': 7, 'reg__max_bin': 136, 'reg__min_data_per_group': 156}

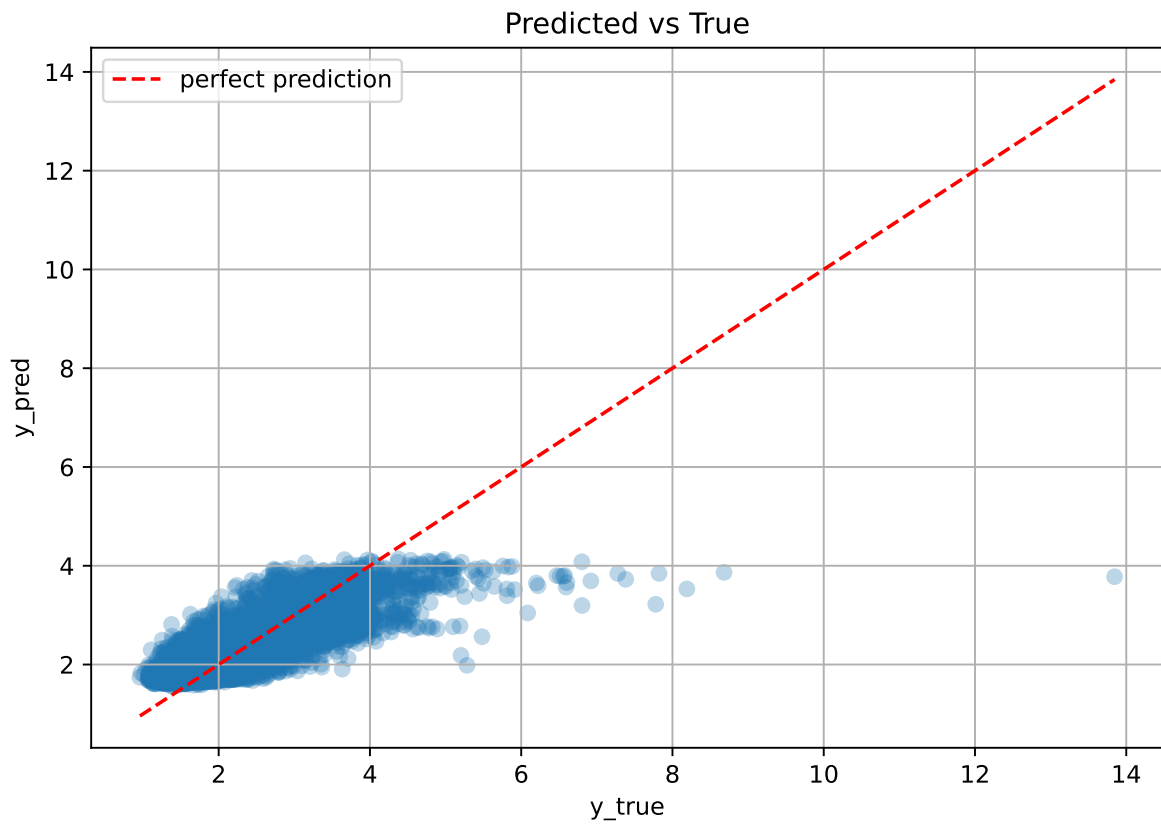
Rank 7: {'reg__n_estimators': 700, 'reg__max_depth': 3, 'reg__learning_rate': 0.021741818640212573, 'reg__num_leaves': 53, 'reg__subsample': 0.7207615058906194, 'reg__colsample_bytree': 0.726303029654088, 'reg__min_child_samples': 17, 'reg__reg_alpha': 8.465110608605018, 'reg__reg_lambda': 31.374613303587772, 'reg__min_split_gain': 3.4045137192699677, 'reg__path_smooth': 4.9882774258810025, 'reg__min_child_weight': 1.7377594591290482, 'reg__feature_fraction': 0.517536774720882, 'reg__bagging_fraction': 0.6278431326514363, 'reg__bagging_freq': 7, 'reg__max_bin': 158, 'reg__min_data_per_group': 133}

Rank 8: {'reg__n_estimators': 300, 'reg__max_depth': 4, 'reg__learning_rate': 0.013812569097104333, 'reg__num_leaves': 26, 'reg__subsample': 0.6761138935313127, 'reg__colsample_bytree': 0.6164144202277362, 'reg__min_child_samples': 15, 'reg__reg_alpha': 0.8891742598581995, 'reg__reg_lambda': 38.28205373780415, 'reg__min_split_gain': 4.348051924249887, 'reg__path_smooth': 3.6785579717075074, 'reg__min_child_weight': 0.12200441679957719, 'reg__feature_fraction': 0.5002923106765346, 'reg__bagging_fraction': 0.545064659044639, 'reg__bagging_freq': 6, 'reg__max_bin': 105, 'reg__min_data_per_group': 145}

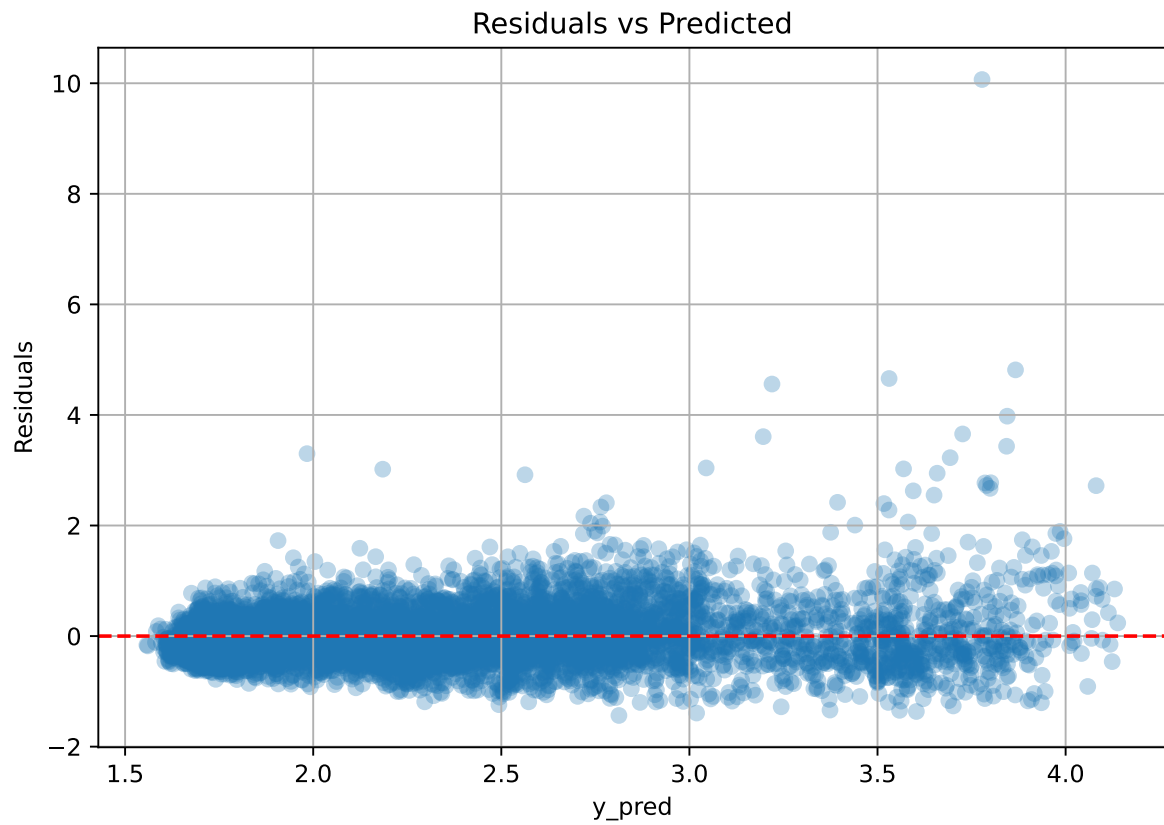
Rank 9: {'reg__n_estimators': 600, 'reg__max_depth': 5, 'reg__learning_rate': 0.016144281365180275, 'reg__num_leaves': 56, 'reg__subsample': 0.7341270868725895, 'reg__colsample_bytree': 0.6188418572113076, 'reg__min_child_samples': 13, 'reg__reg_alpha': 8.107593854502097, 'reg__reg_lambda': 29.09443023560038, 'reg__min_split_gain': 3.566725092451666, 'reg__path_smooth': 4.008938171707776, 'reg__min_child_weight': 0.578318300866888, 'reg__feature_fraction': 0.532737098059274, 'reg__bagging_fraction': 0.5772150731933824, 'reg__bagging_freq': 6, 'reg__max_bin': 136, 'reg__min_data_per_group': 151}

Rank 10: {'reg__n_estimators': 300, 'reg__max_depth': 4, 'reg__learning_rate': 0.012982077502800575, 'reg__num_leaves': 26, 'reg__subsample': 0.6731200323622089, 'reg__colsample_bytree': 0.6901972790746488, 'reg__min_child_samples': 15, 'reg__reg_alpha': 0.8910835881649888, 'reg__reg_lambda': 38.11884904588845, 'reg__min_split_gain': 4.420691630315393, 'reg__path_smooth': 3.8165192463629456, 'reg__min_child_weight': 0.12758683661744916, 'reg__feature_fraction': 0.5007396534117965, 'reg__bagging_fraction': 0.5440344613564451, 'reg__bagging_freq': 5, 'reg__max_bin': 102, 'reg__min_data_per_group': 113}

3.5.2 Scatter



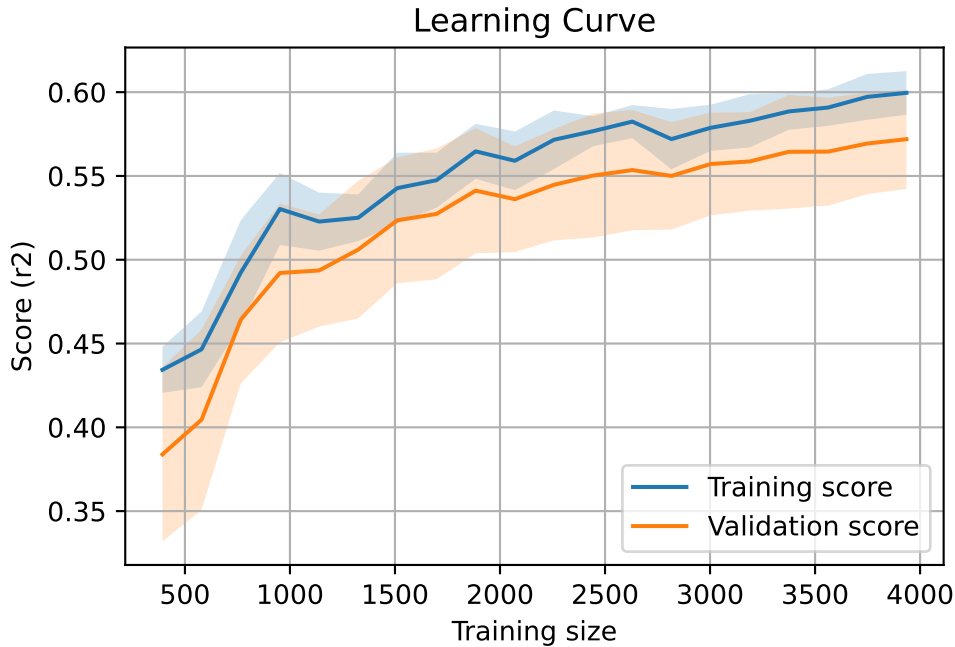
3.5.3 Scatter residuals



3.5.4 Learning curve

`Len(group)` : 8032, `Len(training)` : 5623,

`Len(training_real)` : 3937, `Len(test_of_training)` : 1686, `Len(test)` : 2409



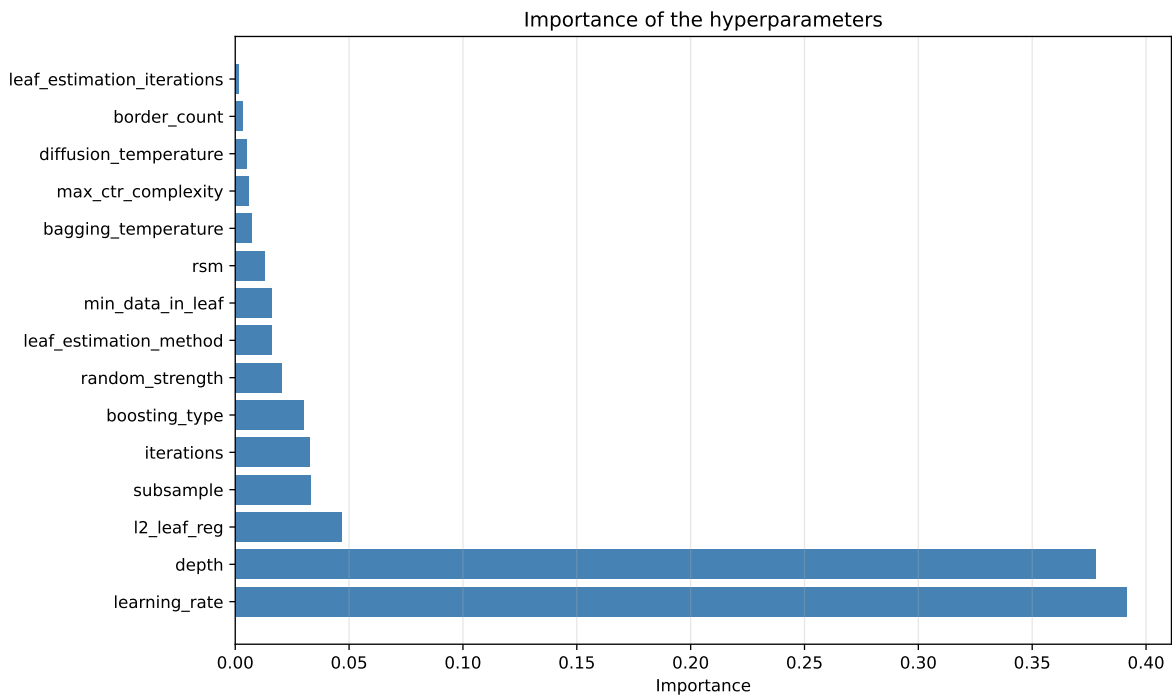
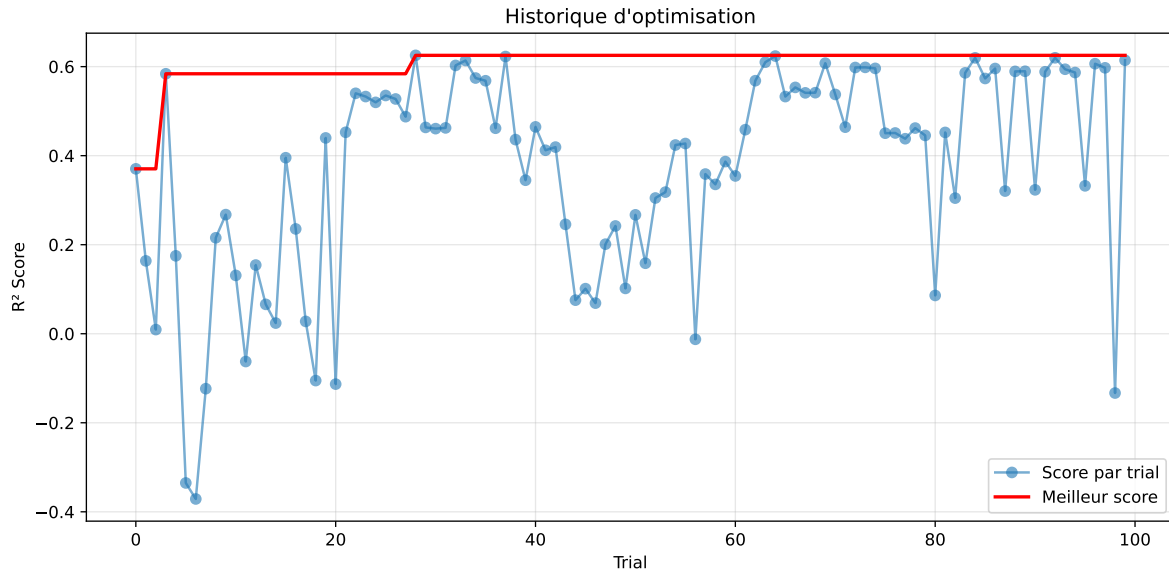
3.6 Model : CatBoost

3.6.1 Gridsearch

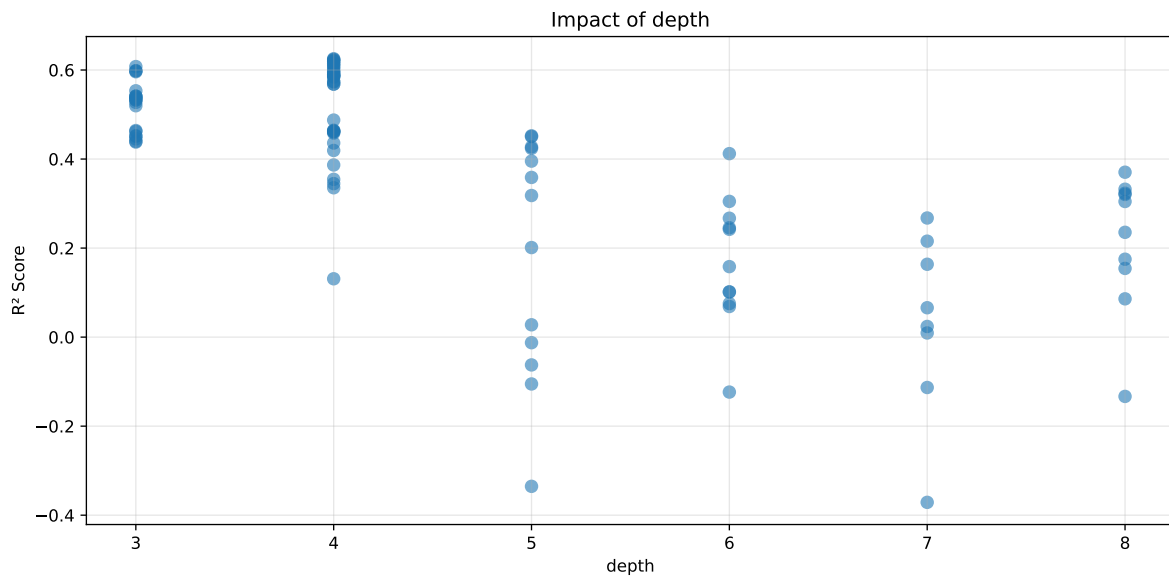
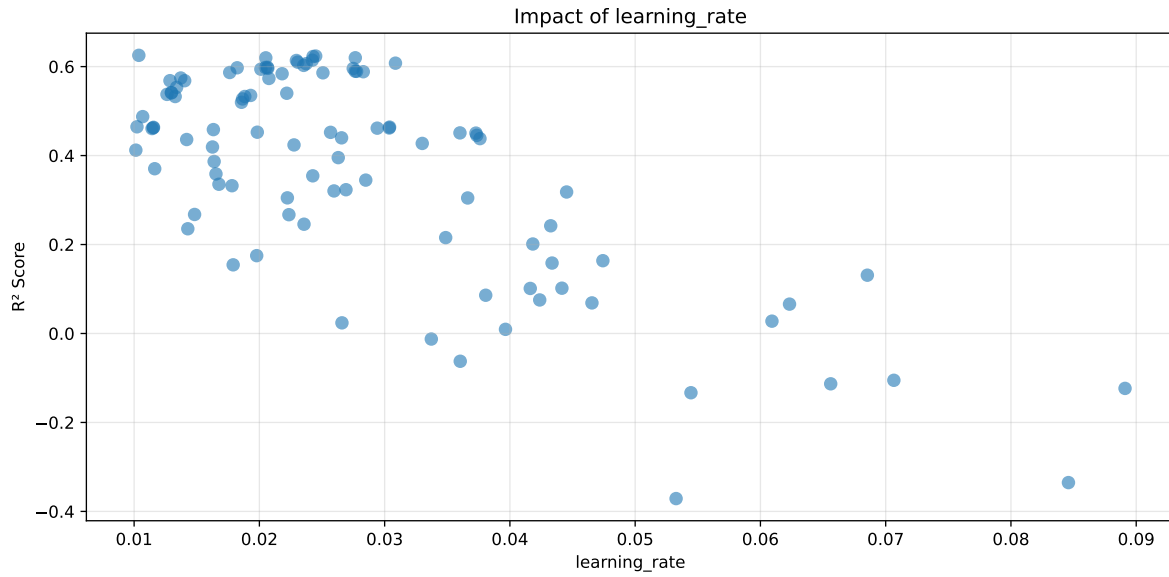
Distribution y_train vs y_test: y_train: mean=2.346, std=0.800, min=0.959, max=13.845
y_test: mean=2.357, std=0.789, min=1.070, max=8.190

===== TOP Importance FEATURES =====

feature	importance
r_temperature_wl_3h_A_rms	23.632698
r_umidity	18.593456
r_temperature_wl_3h_A_energy	10.461398
r_temperature	8.236061
r_CO2	7.394626
r_temperature_wl_3h_D2_energy_rel	4.945661
r_THI	4.413486
r_NH3	4.104038
r_umidity_wl_3h_A_energy_rel	3.480556
r_NH3_wl_3h_D2_energy	3.205880
r_NH3_wl_3h_A_energy	2.623471
r_NH3_wl_3h_D1_energy	2.155481
r_NH3_wl_3h_A_rms	1.992905
r_umidity_wl_3h_D2_energy	1.762374
r_NH3_wl_3h_D2_energy_rel	1.734183
r_NH3_wl_3h_D1_energy_rel	1.263726

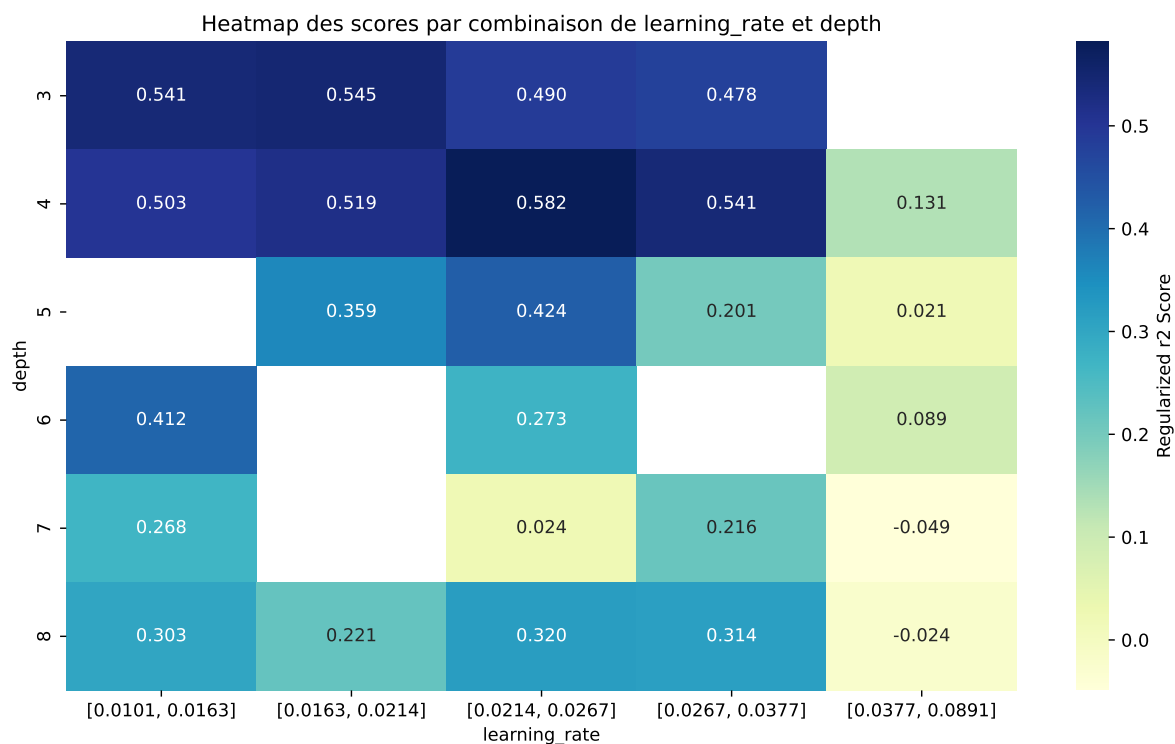


Paramètres avec >5% d'importance : ['learning_rate', 'depth']



C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6558	0.6253	0.6492	0.0308	0.6253	0.6532	0.0279	1.0445	1.0487
2	0.6545	0.6237	0.6464	0.0313	0.6237	0.655	0.0313	1.0501	1.0493
3	0.6537	0.6227	0.6449	0.0316	0.6227	0.6507	0.028	1.0449	1.0497
4	0.649	0.6199	0.6433	0.0324	0.6199	0.6445	0.0246	1.0397	1.0471
5	0.649	0.6197	0.6416	0.0307	0.6197	0.6494	0.0297	1.048	1.0473
6	0.6431	0.6143	0.637	0.0297	0.6143	0.6444	0.0301	1.0489	1.0468
7	0.642	0.6137	0.6359	0.0325	0.6137	0.6454	0.0317	1.0516	1.0461
8	0.6377	0.61	0.631	0.0319	0.61	0.6412	0.0312	1.0511	1.0454
9	0.6331	0.6077	0.6317	0.0307	0.6077	0.6329	0.0252	1.0415	1.0417
10	0.6336	0.6066	0.6303	0.031	0.6066	0.6348	0.0282	1.0465	1.0445
---	---	---	---	---	---	---	---	---	---
100	0.9748	0.7505	0.7847	0.0348	-0.3712	0.7969	0.0464	1.0618	1.2989

Hyperparameters:

Rank 1: {'reg__iterations': 800, 'reg__depth': 4, 'reg__learning_rate': 0.010375954304280463, 'reg__l2_leaf_reg': 35.63118353095352, 'reg__bagging_temperature': 2.9381041058554676, 'reg__random_strength': 2.6754073308455992, 'reg__subsample': 0.8901346833676906, 'reg__min_data_in_leaf': 48, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.704783473149927, 'reg__border_count': 55, 'reg__leaf_estimation_method': 'Gradient',

'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 2028.3409364256033}

Rank 2: {'reg__iterations': 300, 'reg__depth': 4, 'reg__learning_rate': 0.024494897539058494, 'reg__l2_leaf_reg': 24.65736157637942, 'reg__bagging_temperature': 1.609355548546239, 'reg__random_strength': 2.5077533717152156, 'reg__subsample': 0.8259413584350136, 'reg__min_data_in_leaf': 46, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.8957996629350573, 'reg__border_count': 116, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 3564.1924352614287}

Rank 3: {'reg__iterations': 300, 'reg__depth': 4, 'reg__learning_rate': 0.024282822701773336, 'reg__l2_leaf_reg': 19.737530225254126, 'reg__bagging_temperature': 1.8366504359193931, 'reg__random_strength': 2.5391203601724457, 'reg__subsample': 0.894454323606143, 'reg__min_data_in_leaf': 15, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.8117676764820262, 'reg__border_count': 90, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 9860.51767413866}

Rank 4: {'reg__iterations': 300, 'reg__depth': 4, 'reg__learning_rate': 0.027657849479885626, 'reg__l2_leaf_reg': 23.020062091452147, 'reg__bagging_temperature': 0.5830264211098288, 'reg__random_strength': 4.538196376866026, 'reg__subsample': 0.7635698151283673, 'reg__min_data_in_leaf': 48, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.8195517194852461, 'reg__border_count': 66, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 8790.790229764803}

Rank 5: {'reg__iterations': 400, 'reg__depth': 4, 'reg__learning_rate': 0.02051188278073486, 'reg__l2_leaf_reg': 11.976421335964659, 'reg__bagging_temperature': 0.5168845139964464, 'reg__random_strength': 4.173475856674266, 'reg__subsample': 0.5004626245058552, 'reg__min_data_in_leaf': 50, 'reg__leaf_estimation_iterations': 10, 'reg__rsm': 0.8799406137975593, 'reg__border_count': 64, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 6916.40297464908}

Rank 6: {'reg__iterations': 300, 'reg__depth': 4, 'reg__learning_rate': 0.024226574533359725, 'reg__l2_leaf_reg': 20.620965682552413, 'reg__bagging_temperature': 4.679381853628392, 'reg__random_strength': 2.961658393754639, 'reg__subsample': 0.6852470290225505, 'reg__min_data_in_leaf': 48, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.8003892129756817, 'reg__border_count': 74, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 5996.481643487411}

Rank 7: {'reg__iterations': 300, 'reg__depth': 4, 'reg__learning_rate': 0.022956003751745876, 'reg__l2_leaf_reg': 24.466964174698052, 'reg__bagging_temperature': 1.7651040239380906, 'reg__random_strength': 2.706233785733768, 'reg__subsample': 0.8279263183795108,

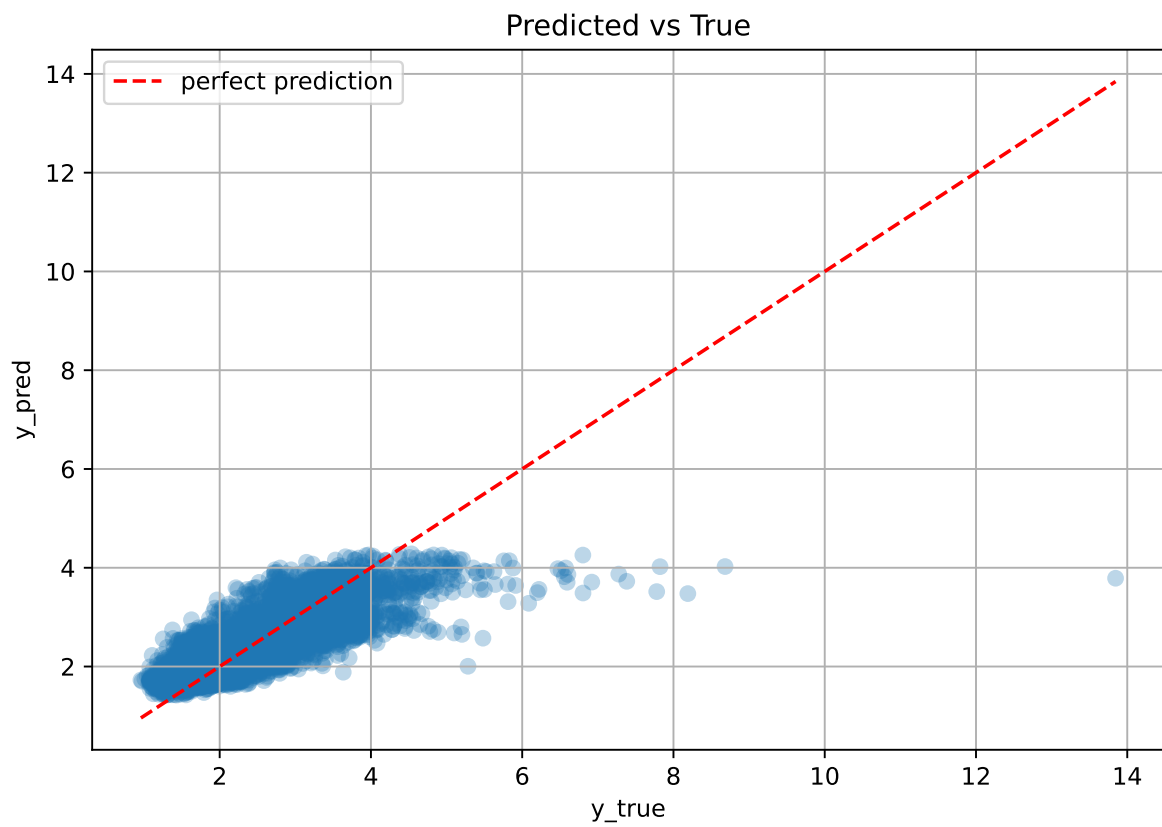
'reg__min_data_in_leaf': 17, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.6767067751433206, 'reg__border_count': 113, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 1952.4714992598333}

Rank 8: {'reg__iterations': 300, 'reg__depth': 4, 'reg__learning_rate': 0.023062747006066295, 'reg__l2_leaf_reg': 25.515826736882097, 'reg__bagging_temperature': 0.7108034650786634, 'reg__random_strength': 3.1980214708384405, 'reg__subsample': 0.8318168066246471, 'reg__min_data_in_leaf': 23, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.8538286125889598, 'reg__border_count': 73, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 3586.1162085723136}

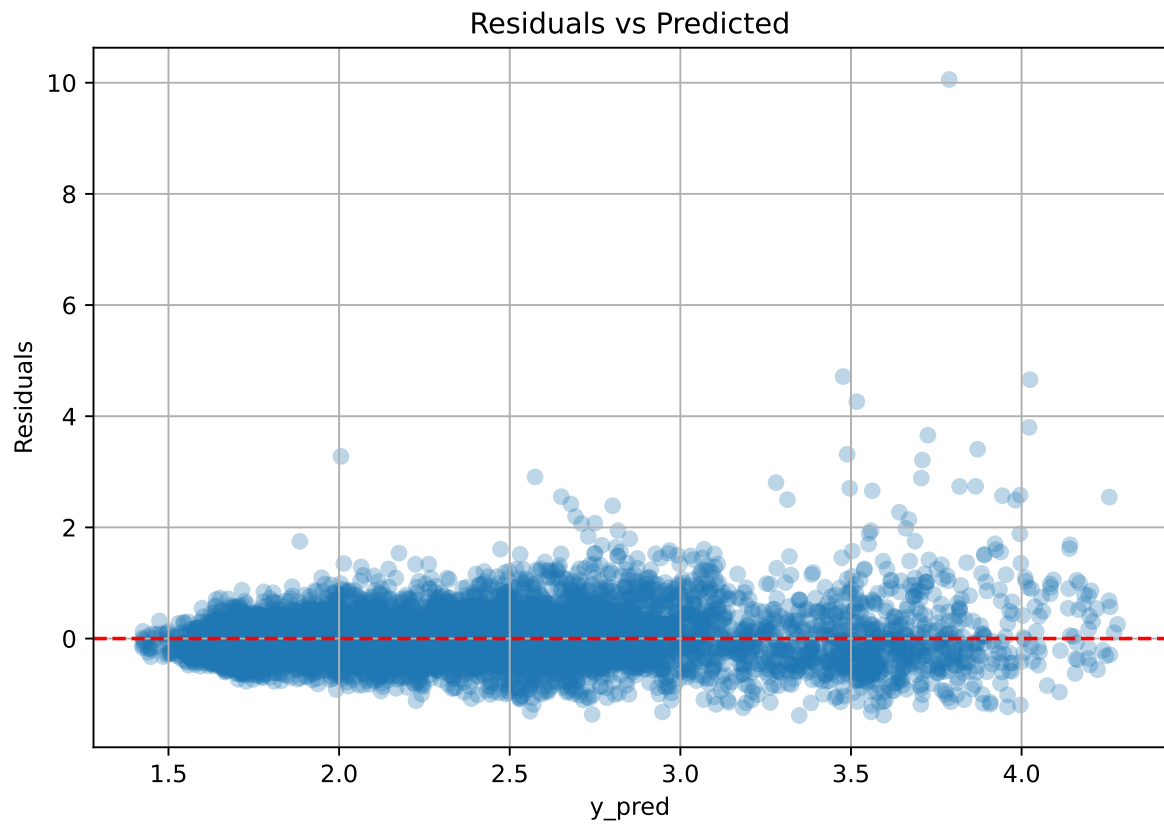
Rank 9: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.030863867309593437, 'reg__l2_leaf_reg': 24.106828259020368, 'reg__bagging_temperature': 4.334206728533841, 'reg__random_strength': 3.6576026696606276, 'reg__subsample': 0.8291010093650636, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.8503107917325174, 'reg__border_count': 115, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 7118.36657906738}

Rank 10: {'reg__iterations': 300, 'reg__depth': 4, 'reg__learning_rate': 0.023736023132240783, 'reg__l2_leaf_reg': 20.55194469845402, 'reg__bagging_temperature': 2.409417721321491, 'reg__random_strength': 4.528494108859214, 'reg__subsample': 0.5558080560107732, 'reg__min_data_in_leaf': 17, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.8202554786765315, 'reg__border_count': 60, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 9108.531138149265}

3.6.2 Scatter



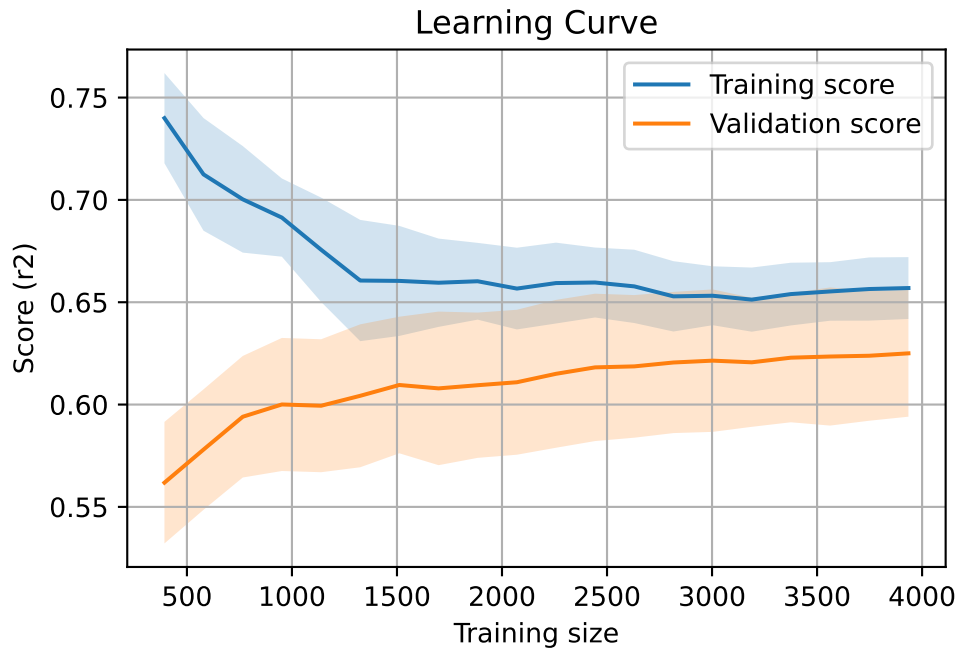
3.6.3 Scatter residuals



3.6.4 Learning curve

`Len(group)` : 8032, `Len(training)` : 5623,

`Len(training_real)` : 3937, `Len(test_of_training)` : 1686, `Len(test)` : 2409



3.7 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.539	0.529	0.033	0.561	0.562	0.032	1.061	1.019	nan
LGBM	0.602	0.575	0.031	0.596	0.62	0.045	1.078	1.047	0.5753
CatBoost	0.656	0.625	0.031	0.649	0.653	0.028	1.044	1.049	0.6253

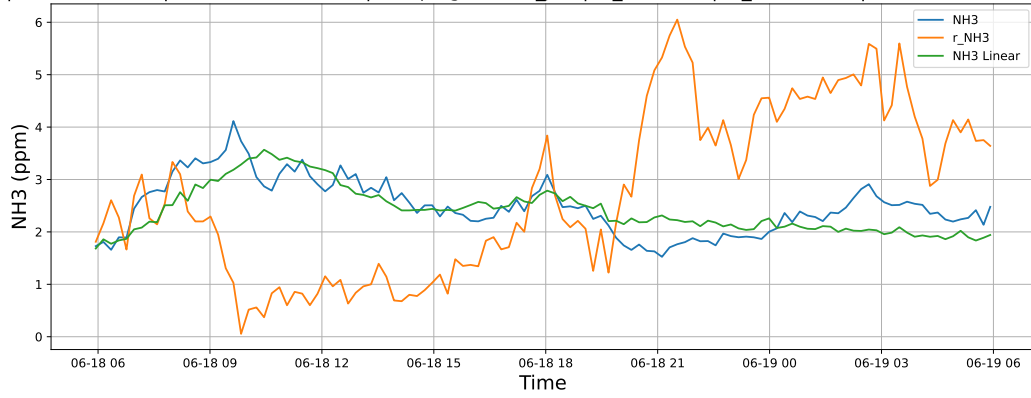
4 Plot

4.1 Standalone plots

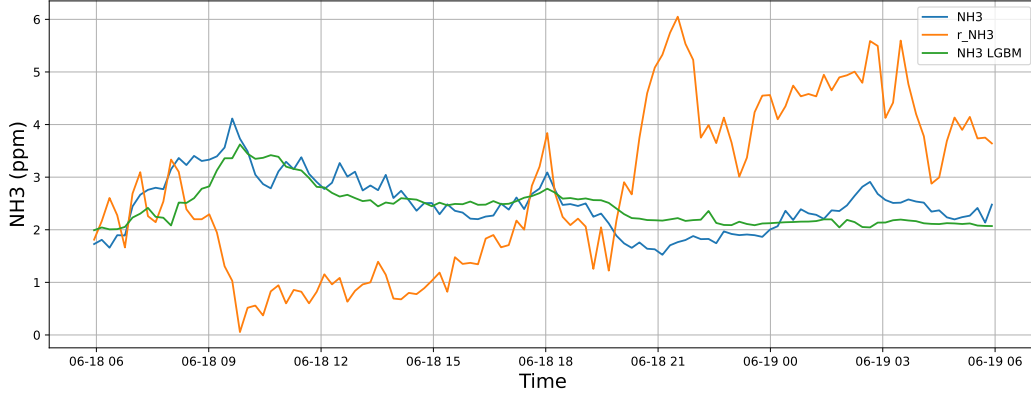
4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

4.1.1.1 Time window : 24

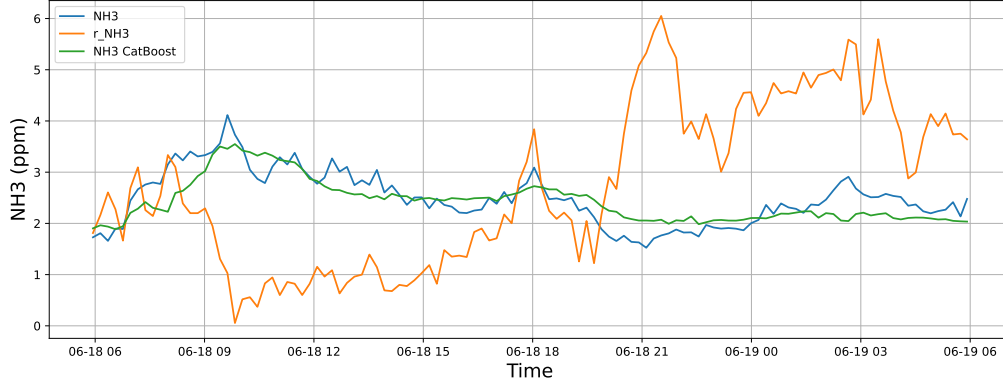
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



4.1.1.2 Time window : 48

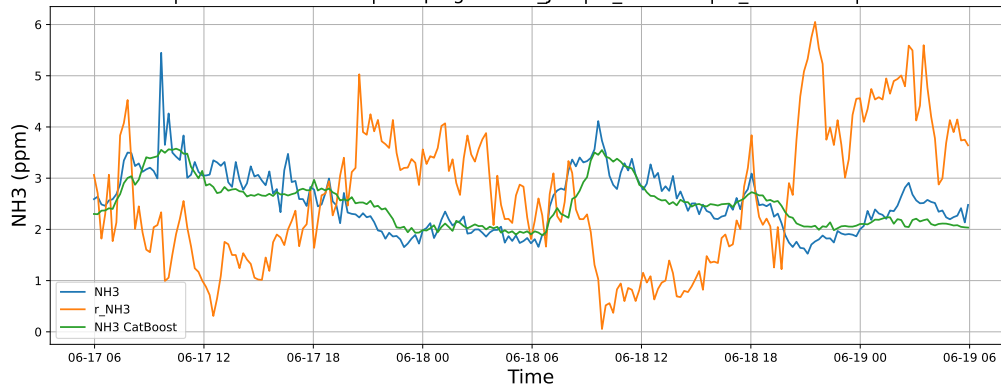
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

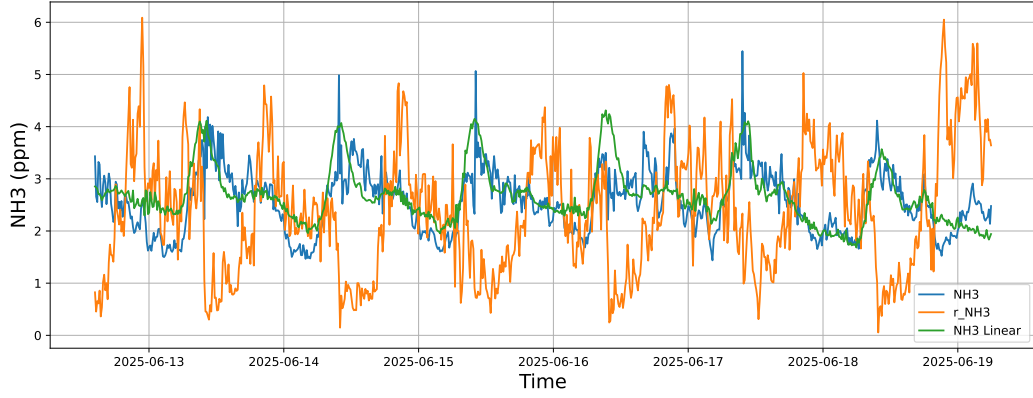


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

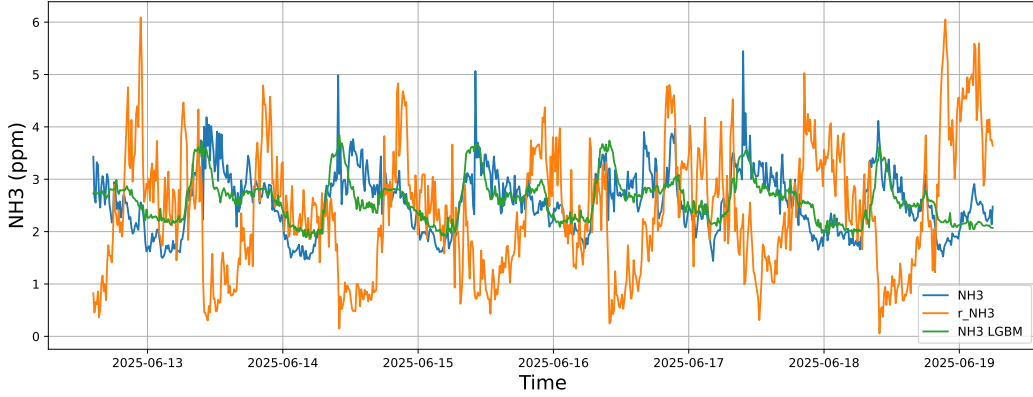


4.1.1.3 Time window : ALL

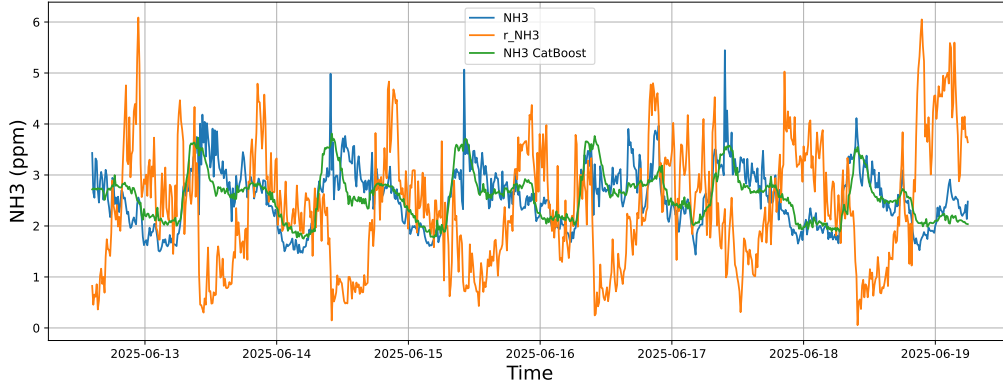
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



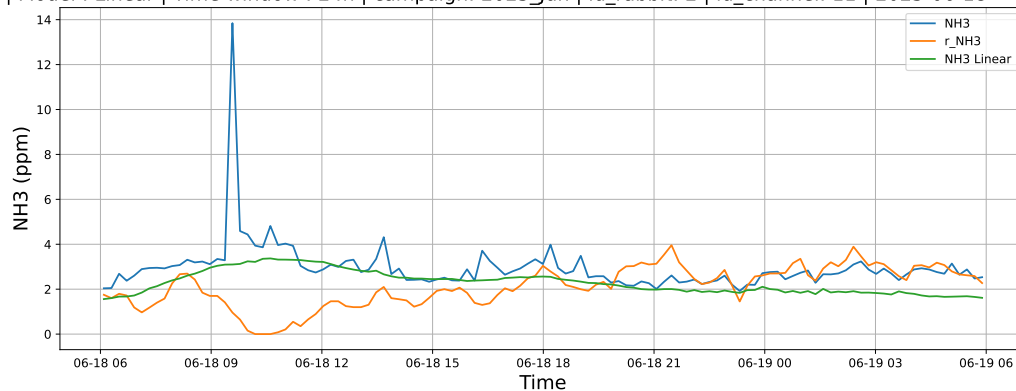
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



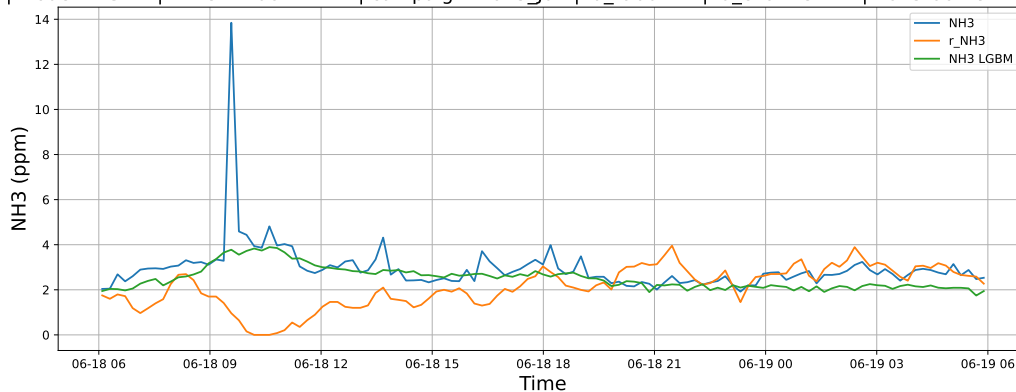
4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.1.2.1 Time window : 24

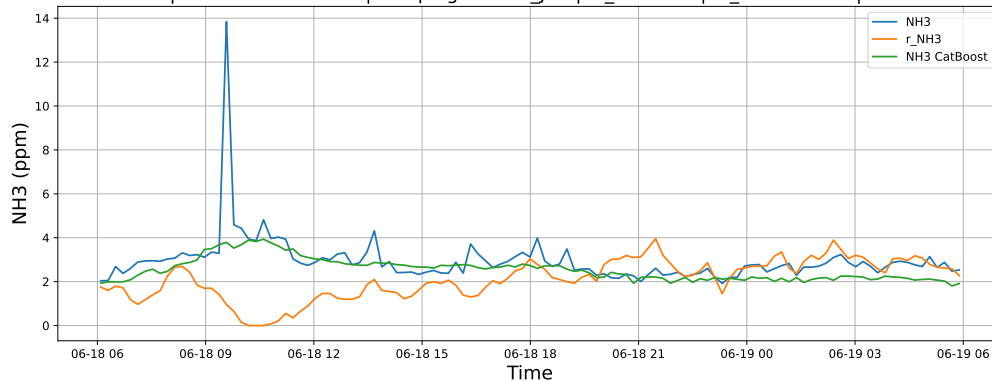
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

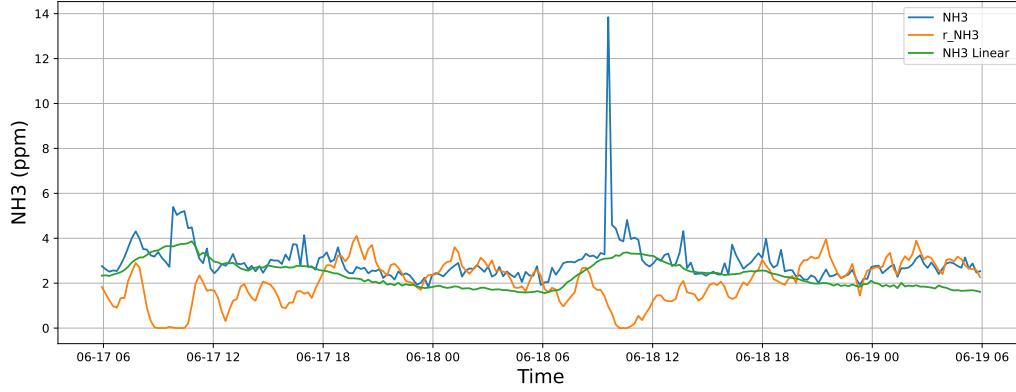


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

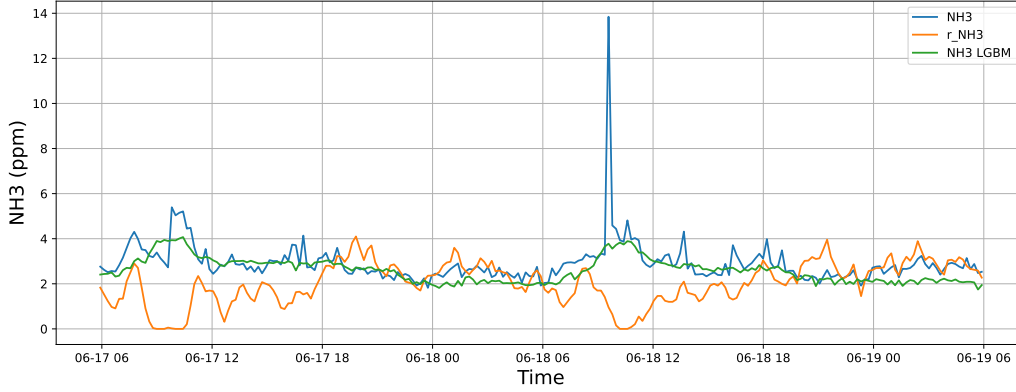


4.1.2.2 Time window : 48

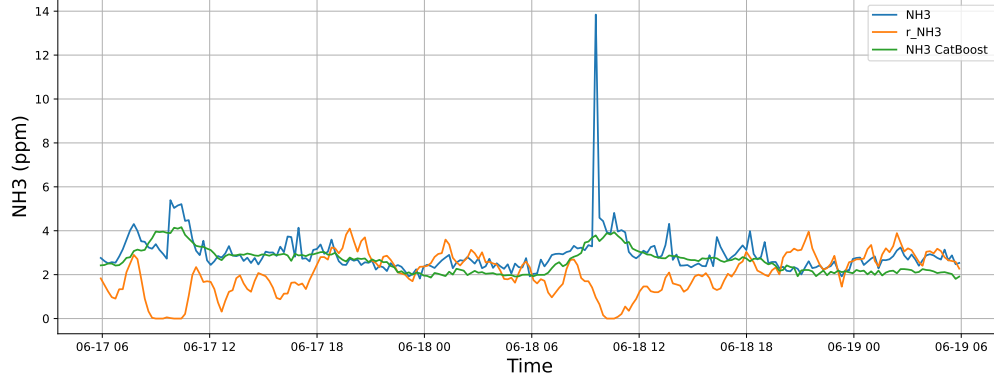
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

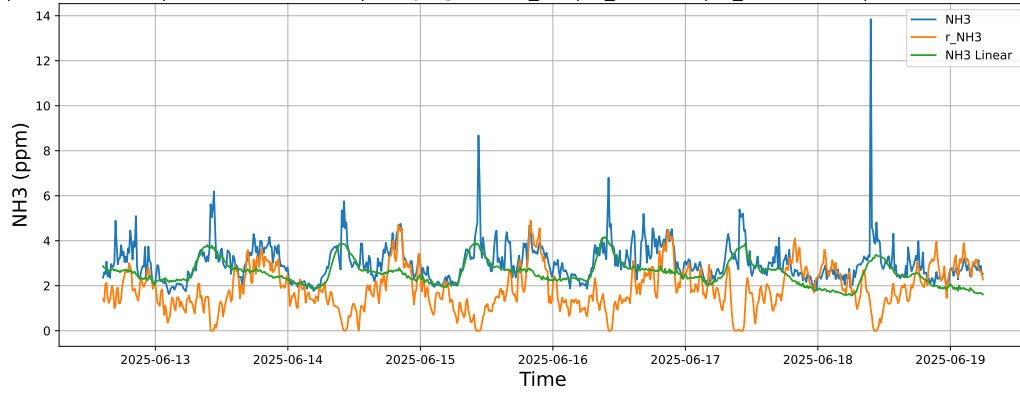


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

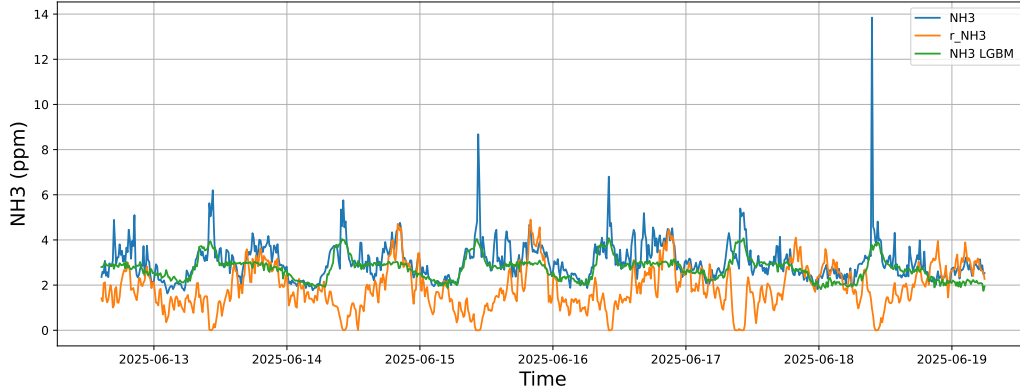


4.1.2.3 Time window : ALL

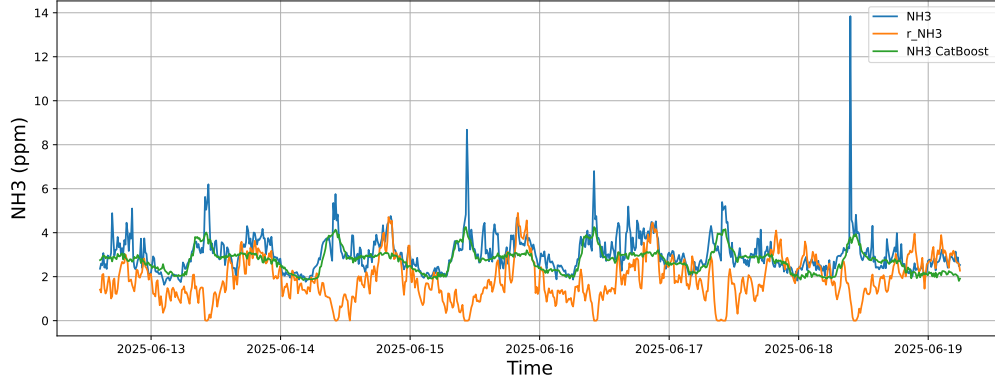
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



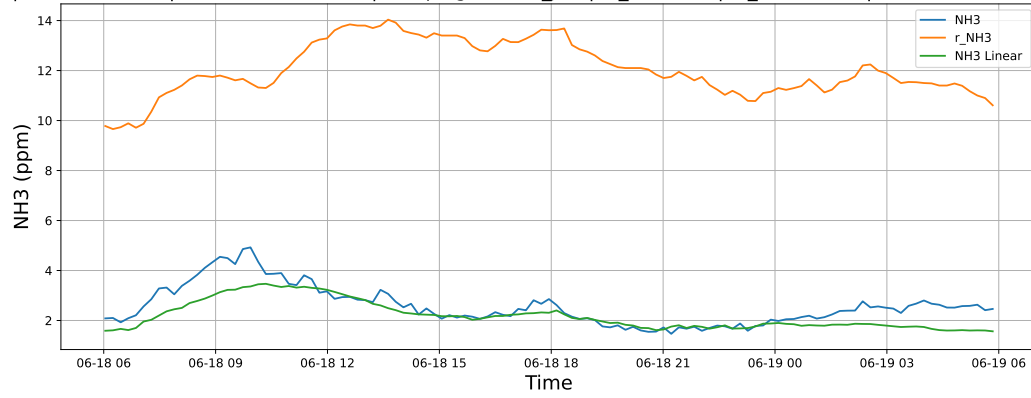
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



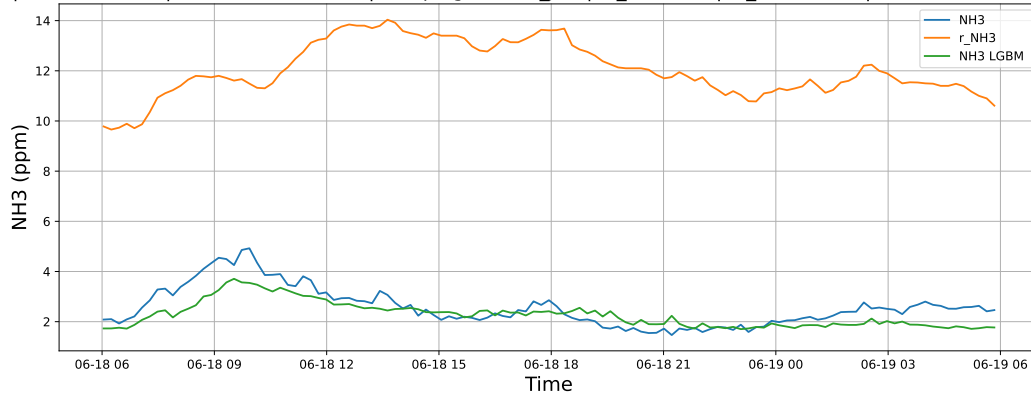
4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.1.3.1 Time window : 24

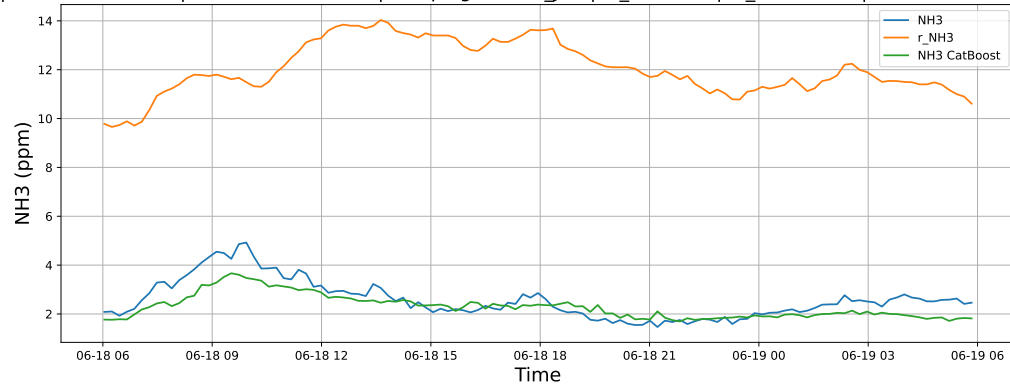
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

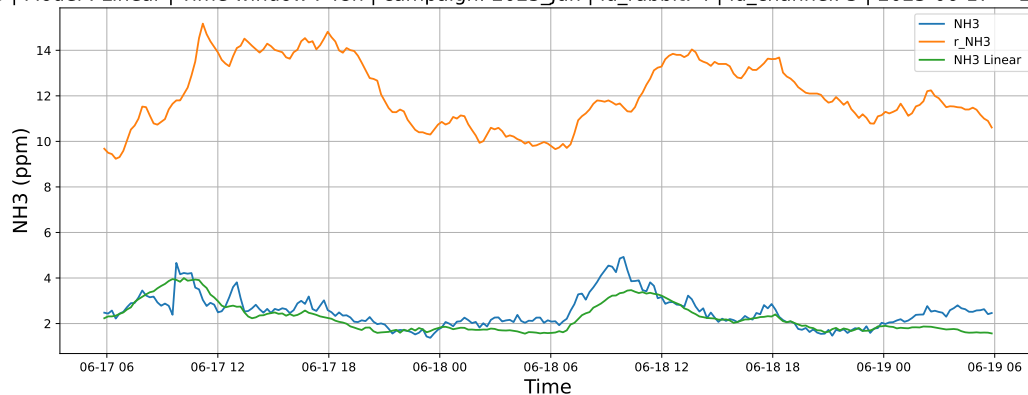


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

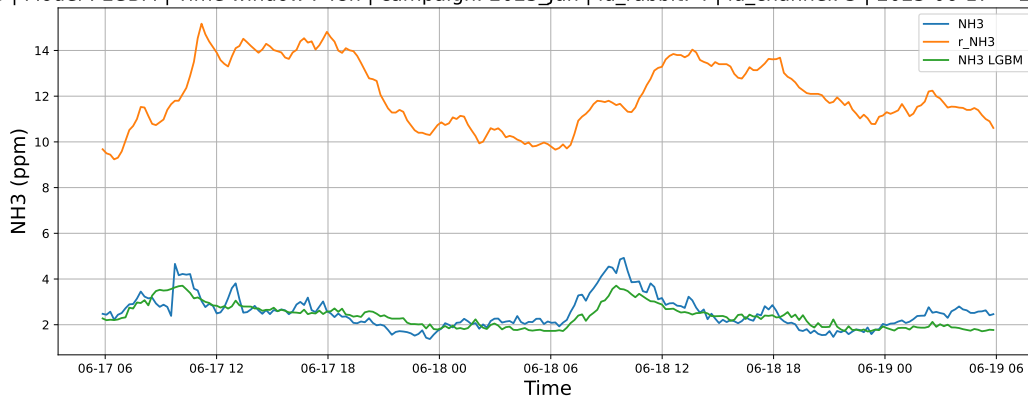


4.1.3.2 Time window : 48

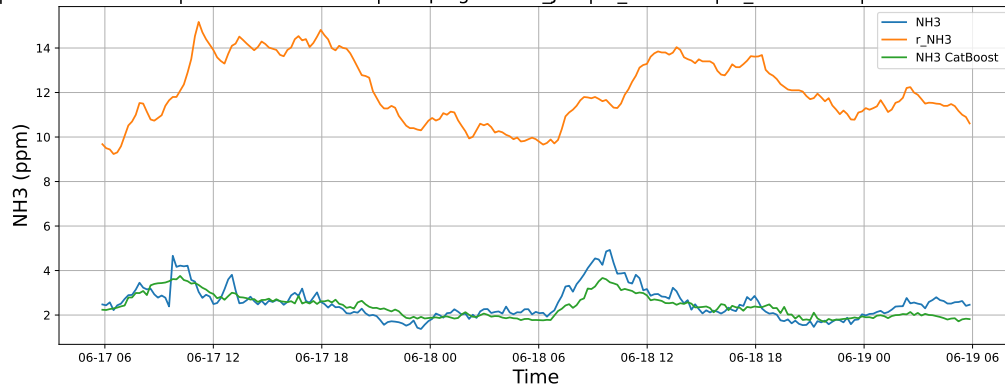
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

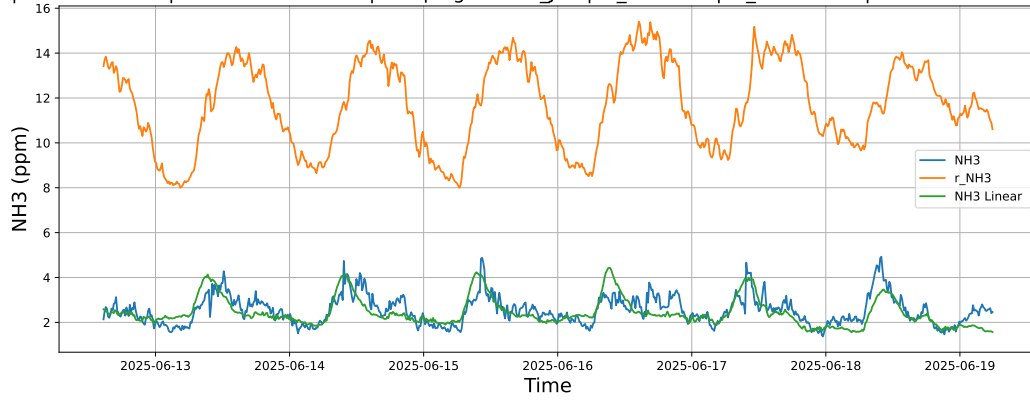


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

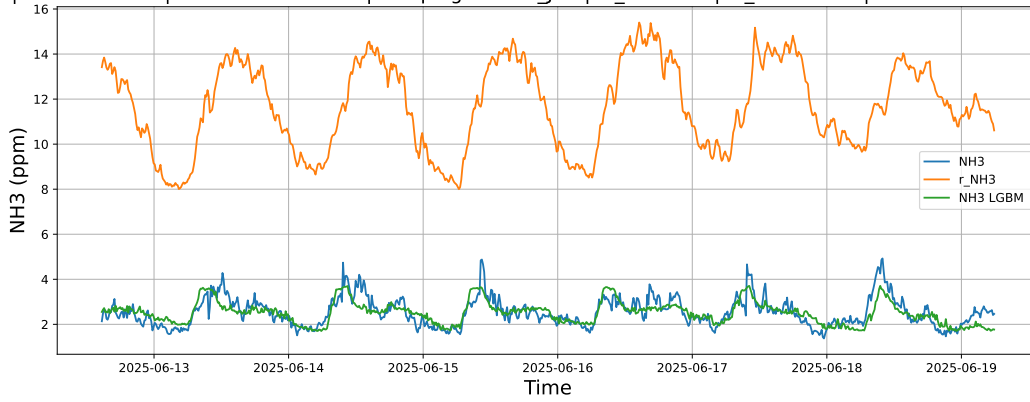


4.1.3.3 Time window : ALL

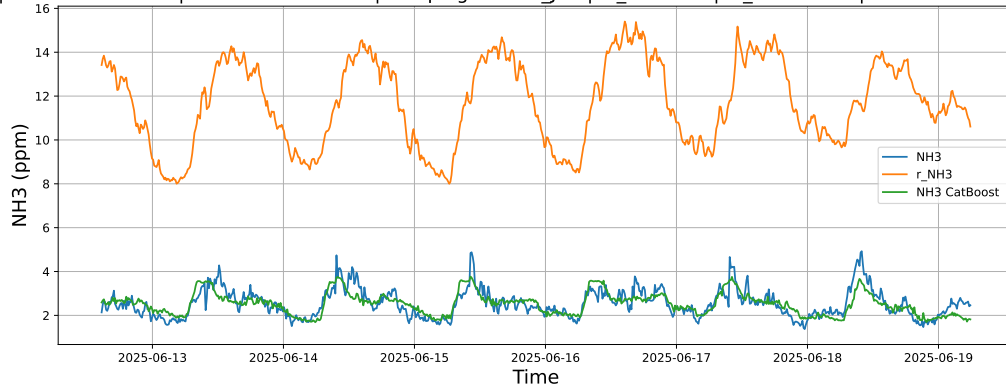
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



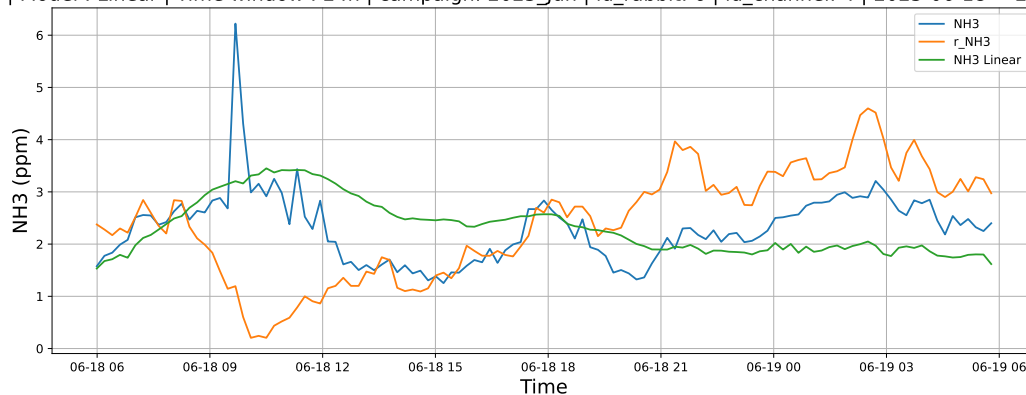
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



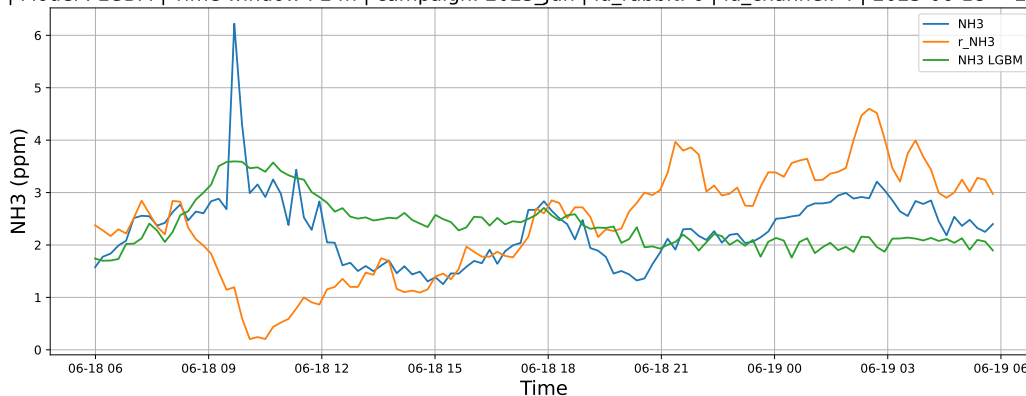
4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

4.1.4.1 Time window : 24

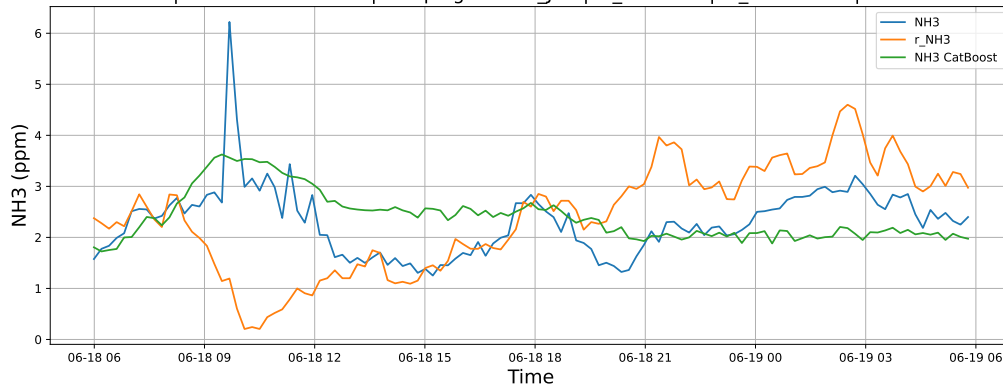
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

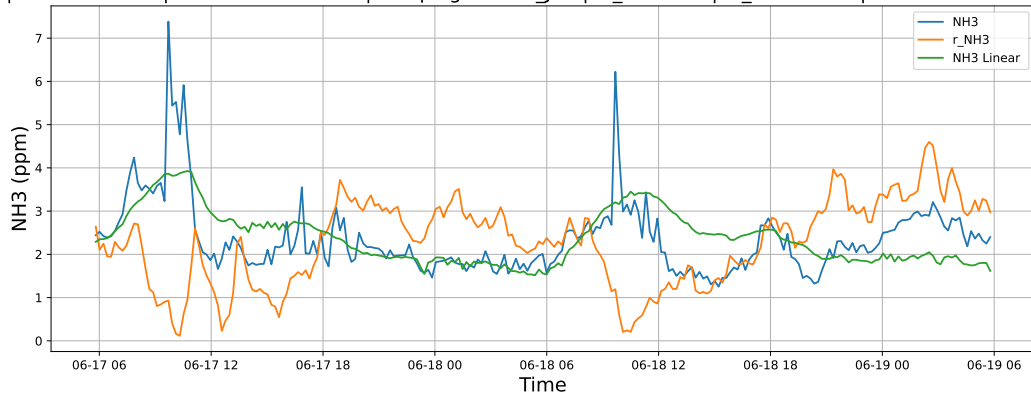


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

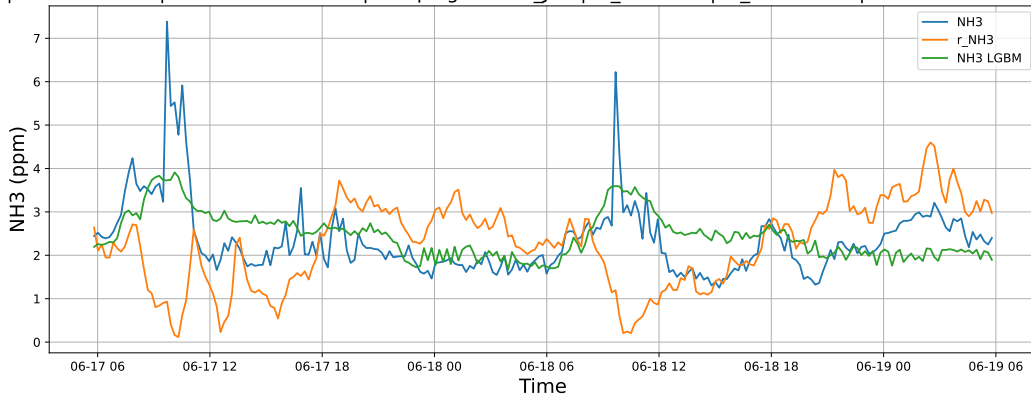


4.1.4.2 Time window : 48

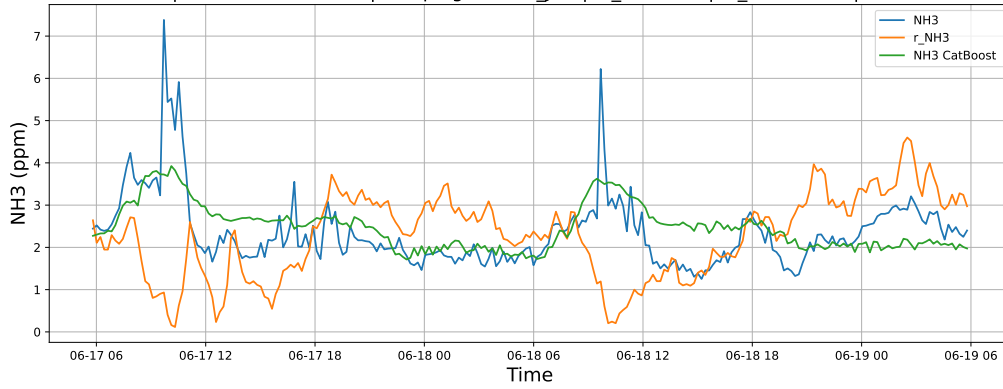
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

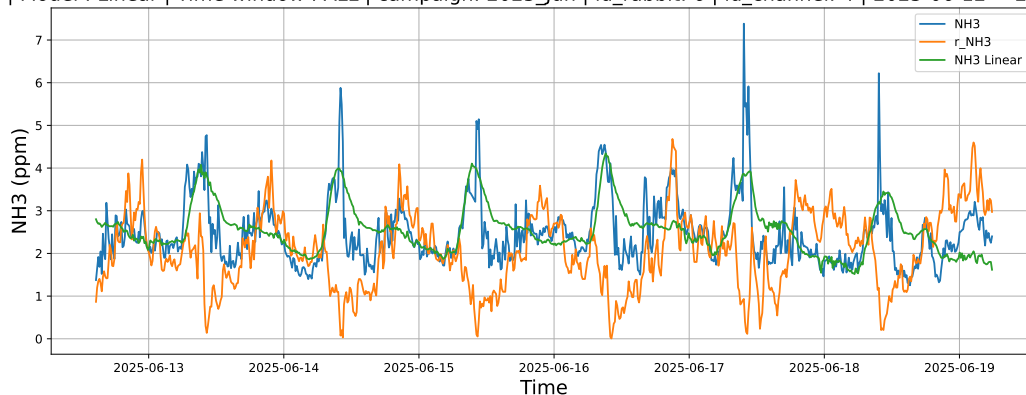


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

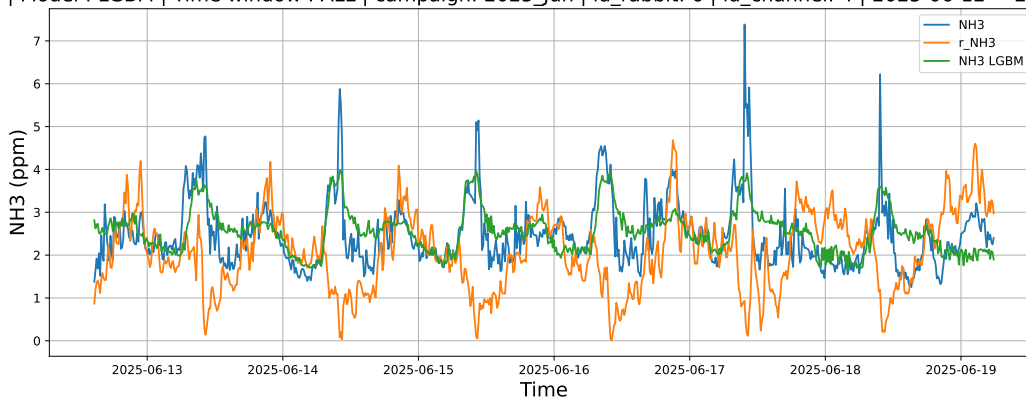


4.1.4.3 Time window : ALL

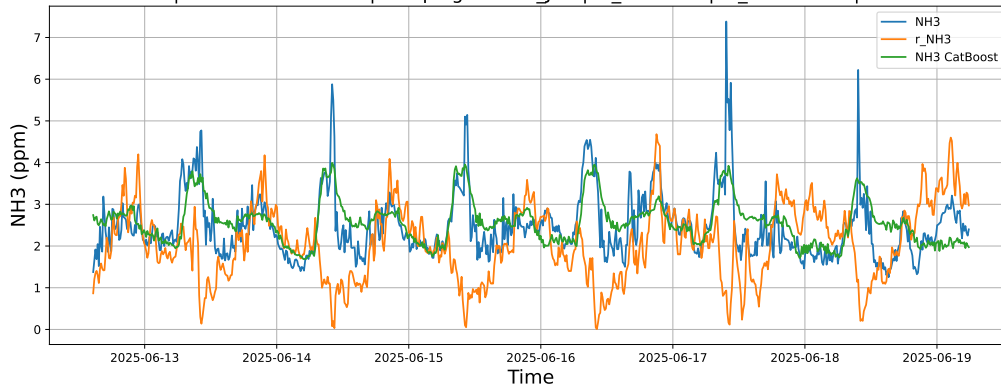
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



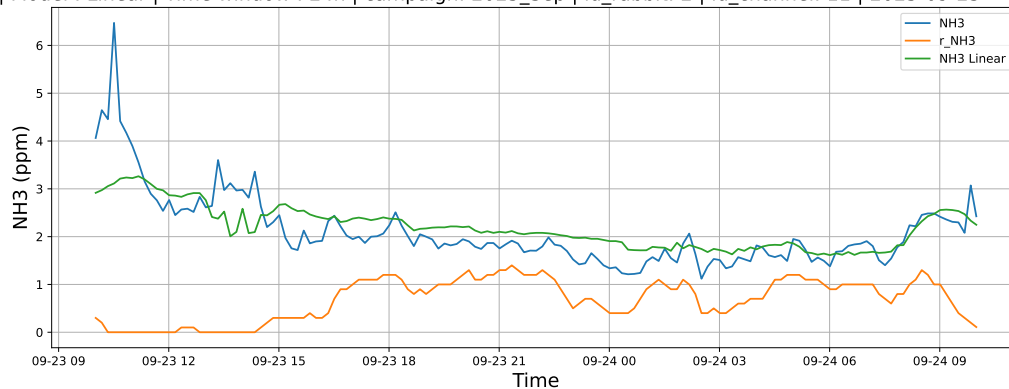
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



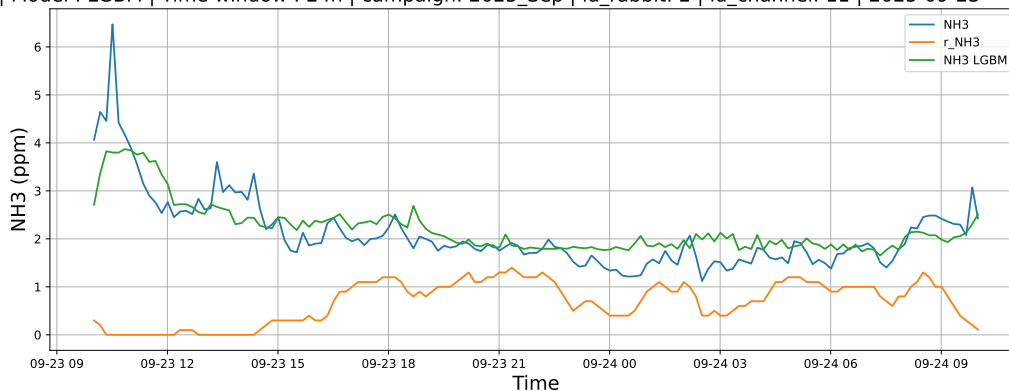
4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

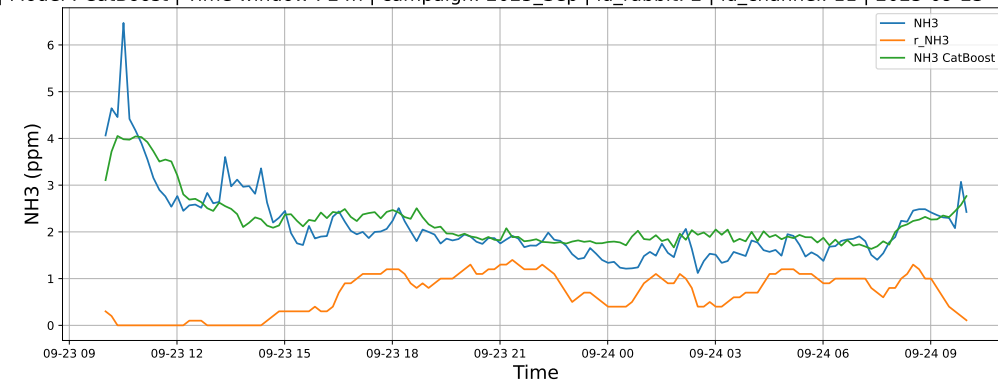
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

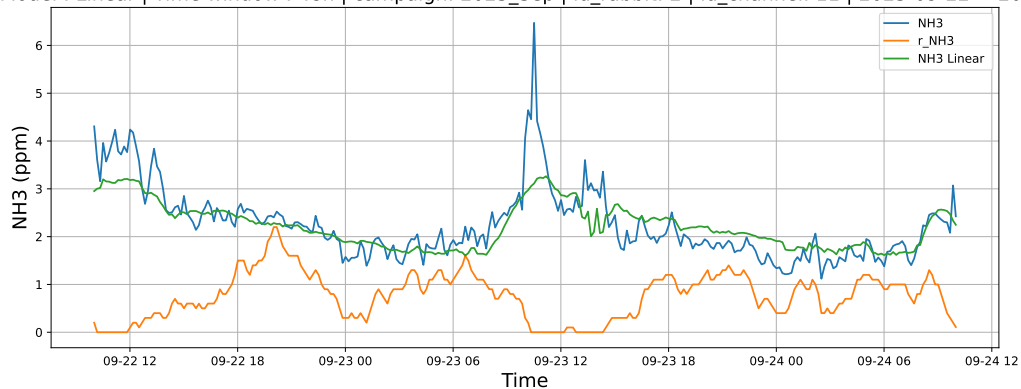


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

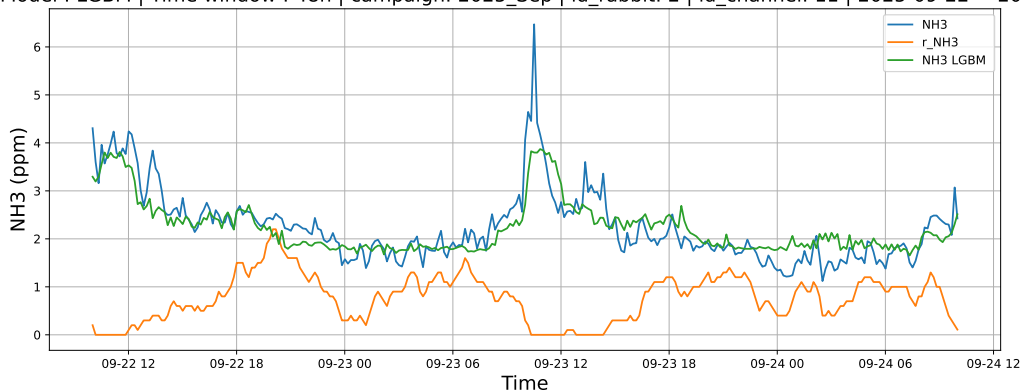


4.1.5.2 Time window : 48

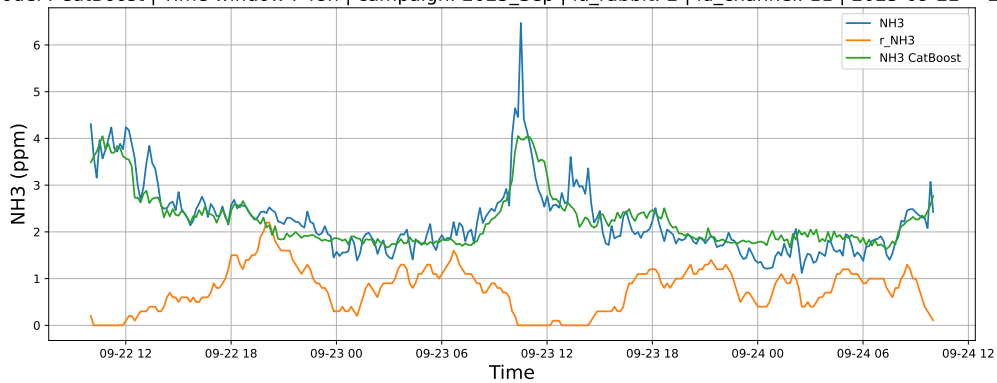
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

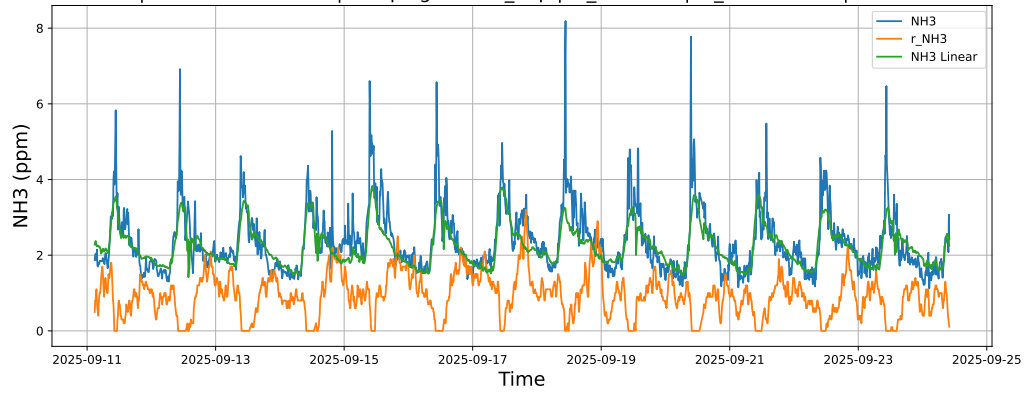


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

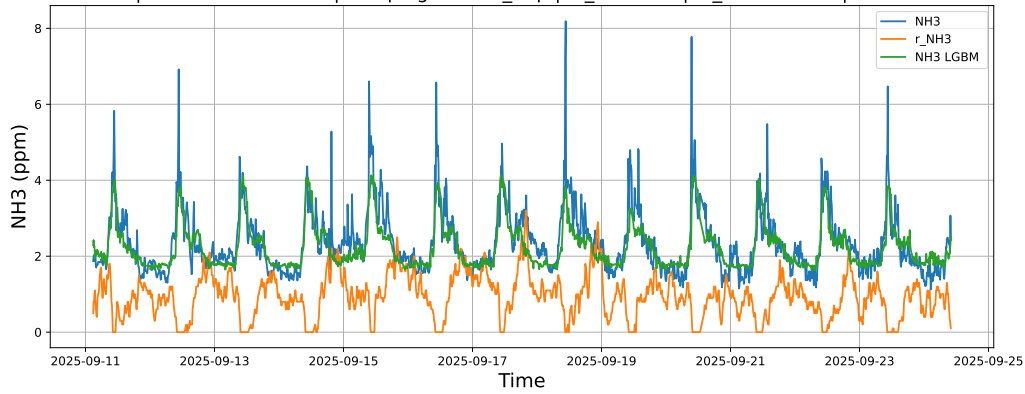


4.1.5.3 Time window : ALL

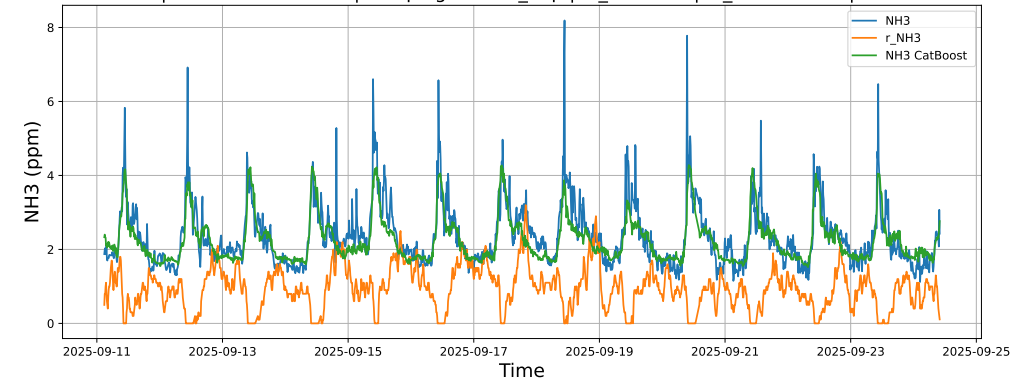
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



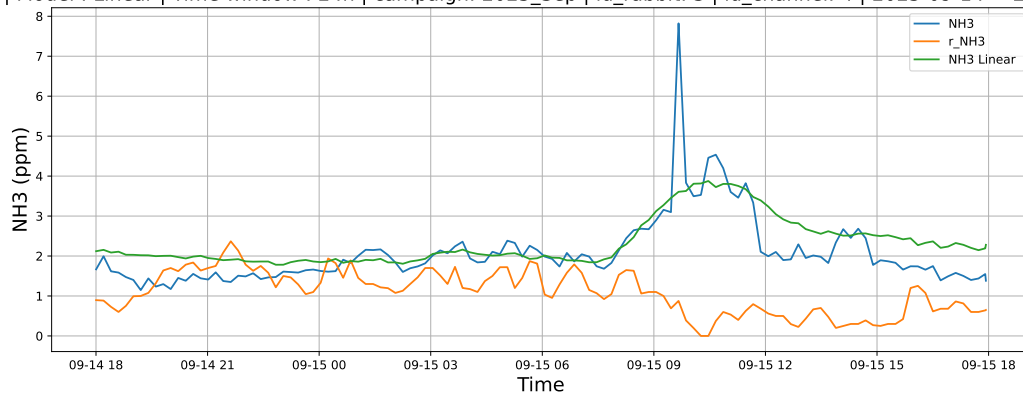
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



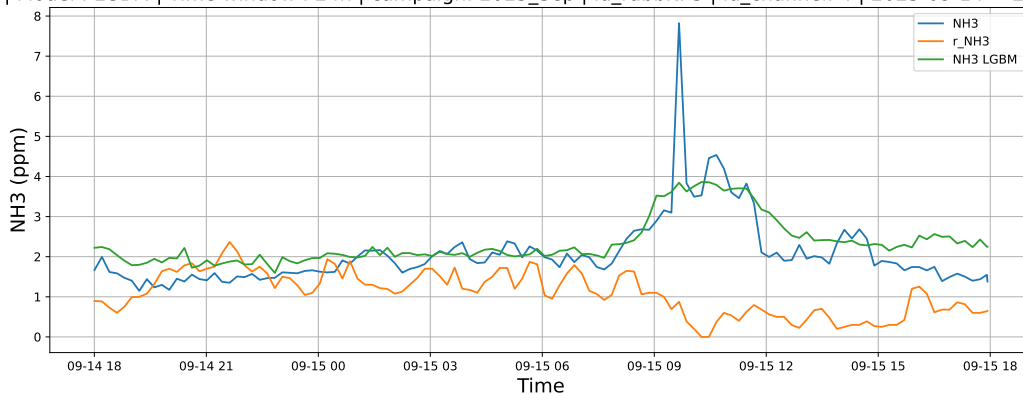
4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.1.6.1 Time window : 24

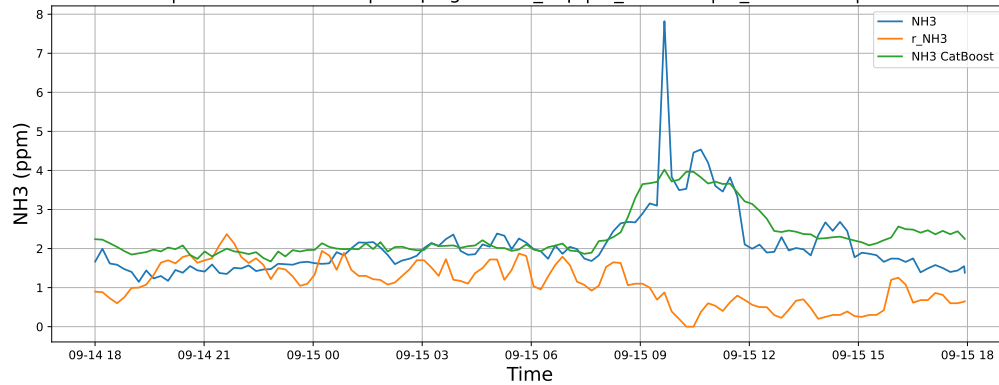
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

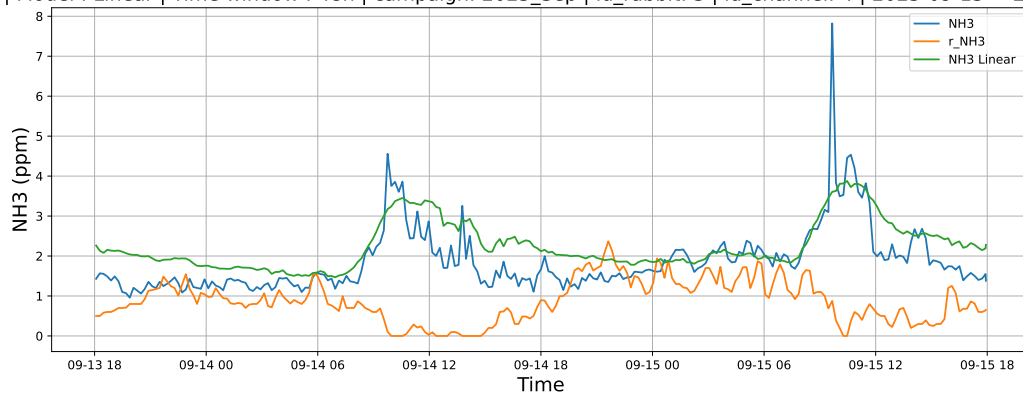


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

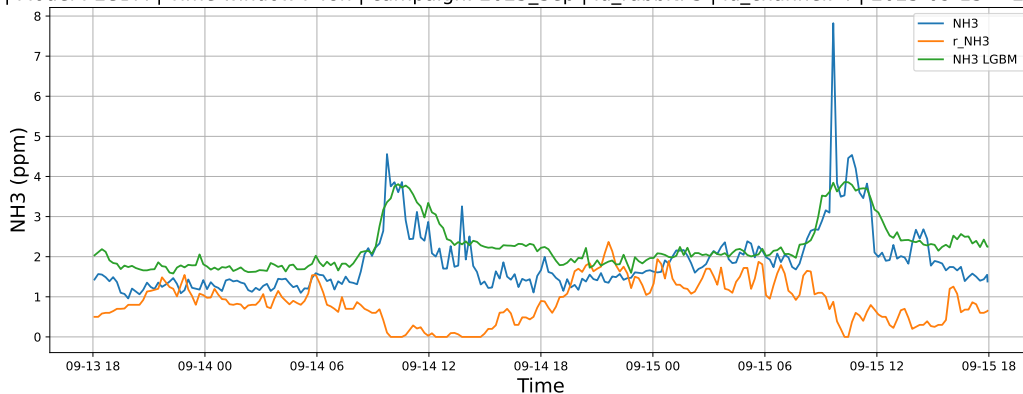


4.1.6.2 Time window : 48

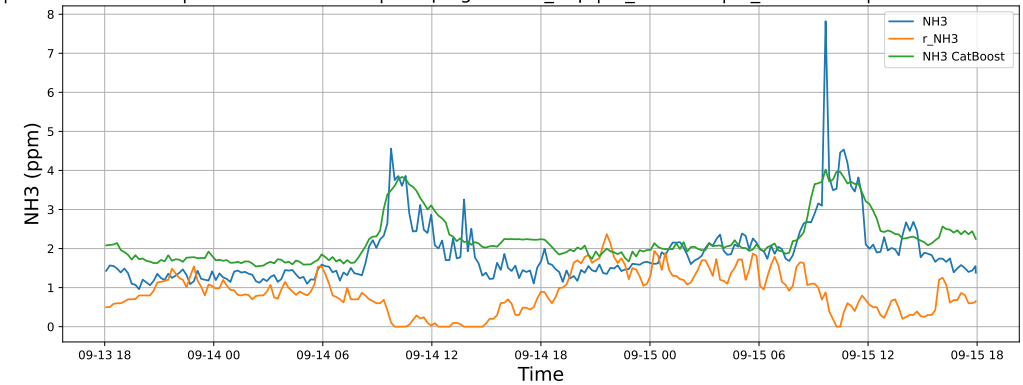
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

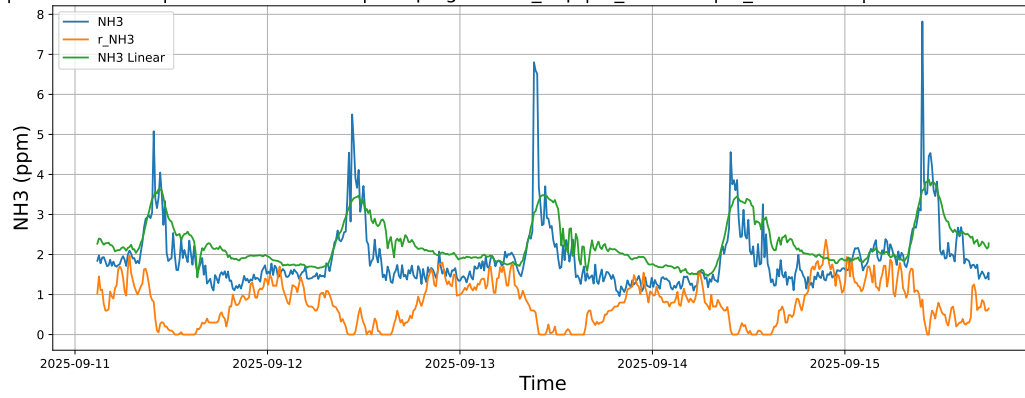


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

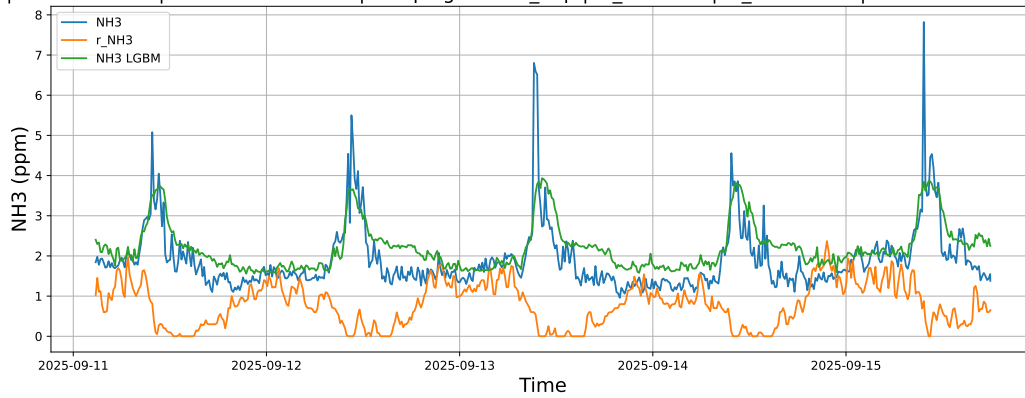


4.1.6.3 Time window : ALL

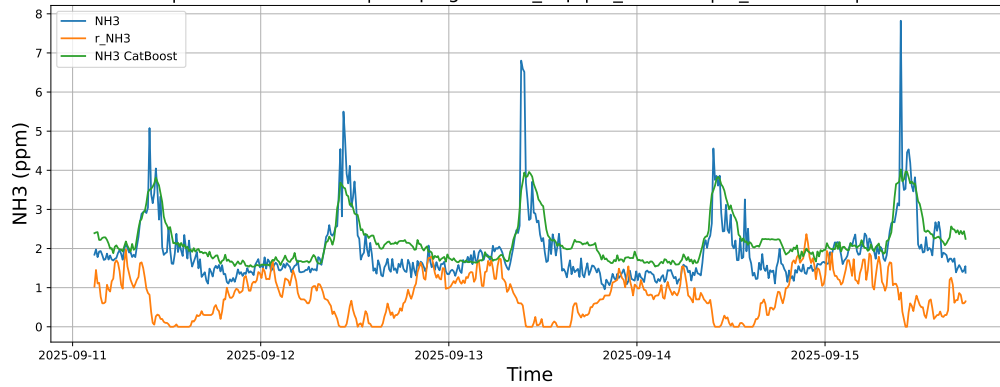
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



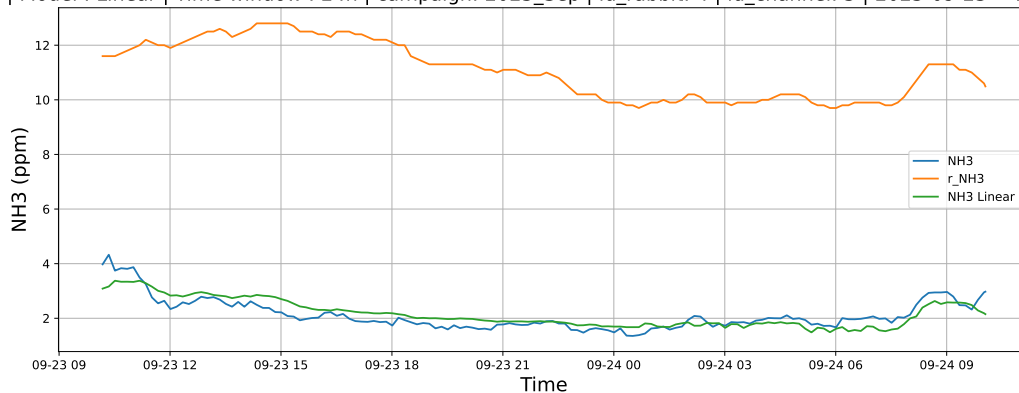
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



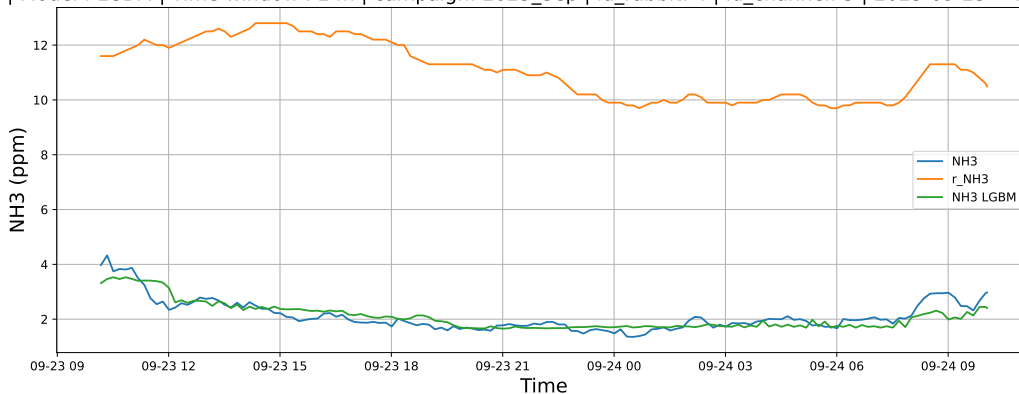
4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

4.1.7.1 Time window : 24

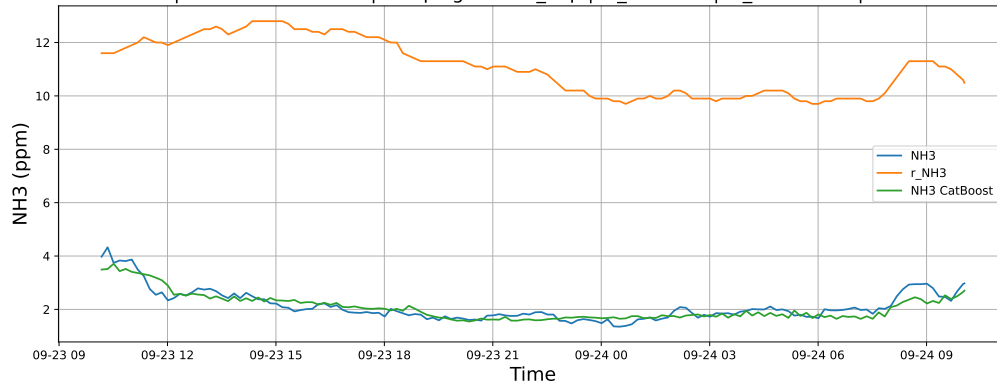
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

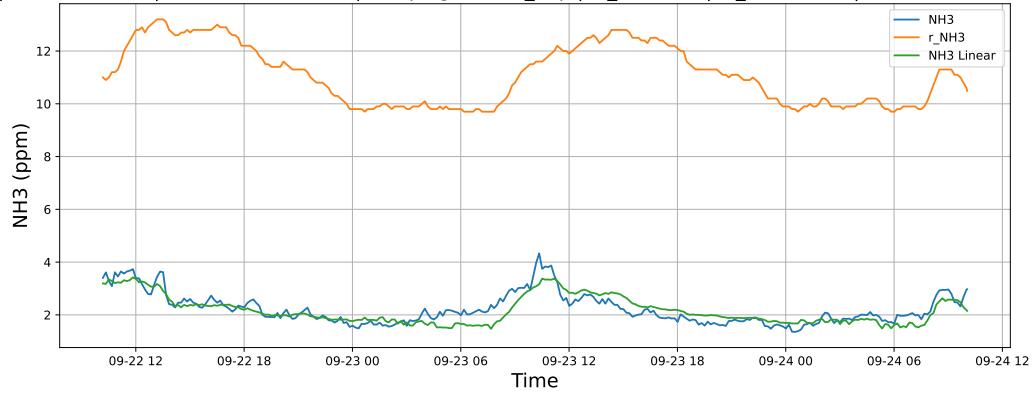


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

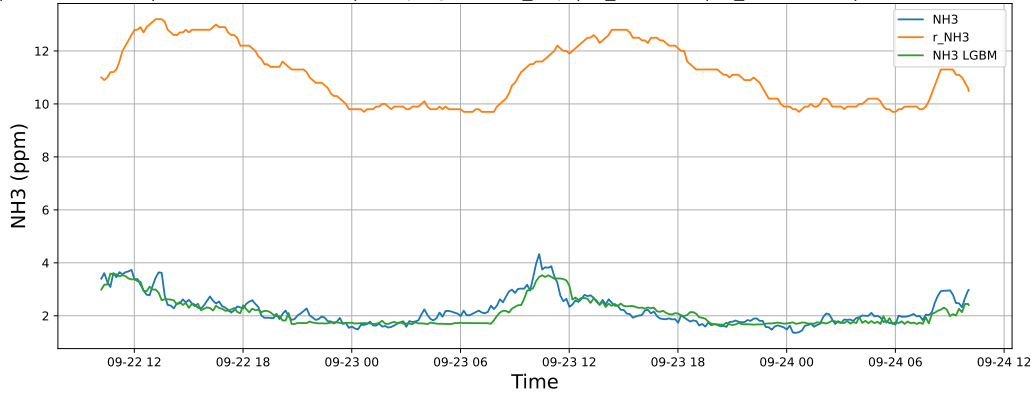


4.1.7.2 Time window : 48

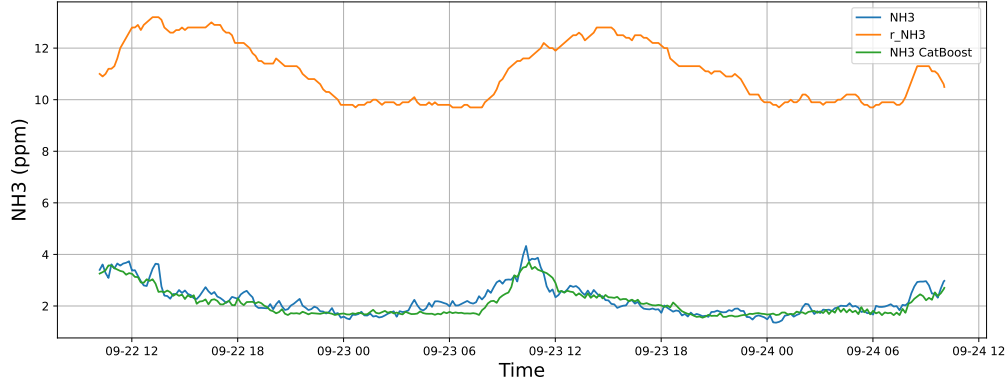
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

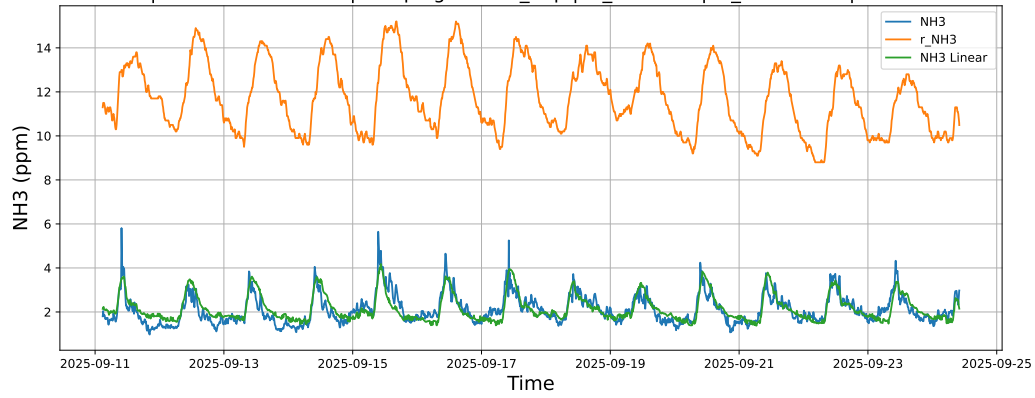


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

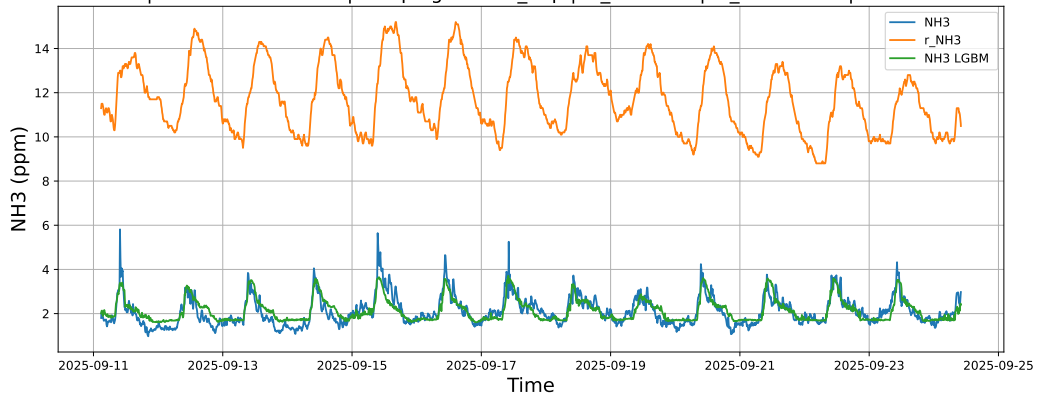


4.1.7.3 Time window : ALL

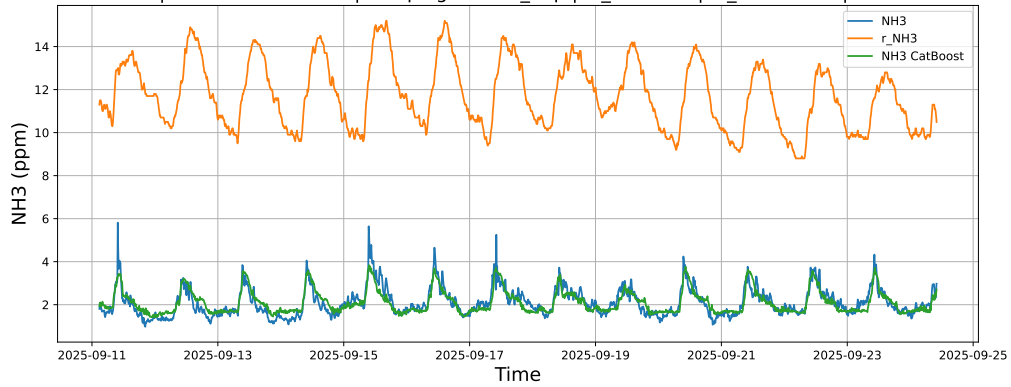
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



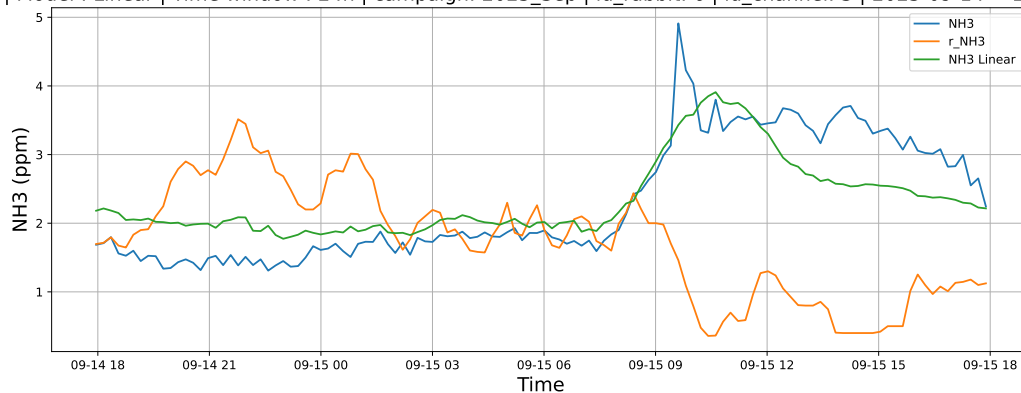
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



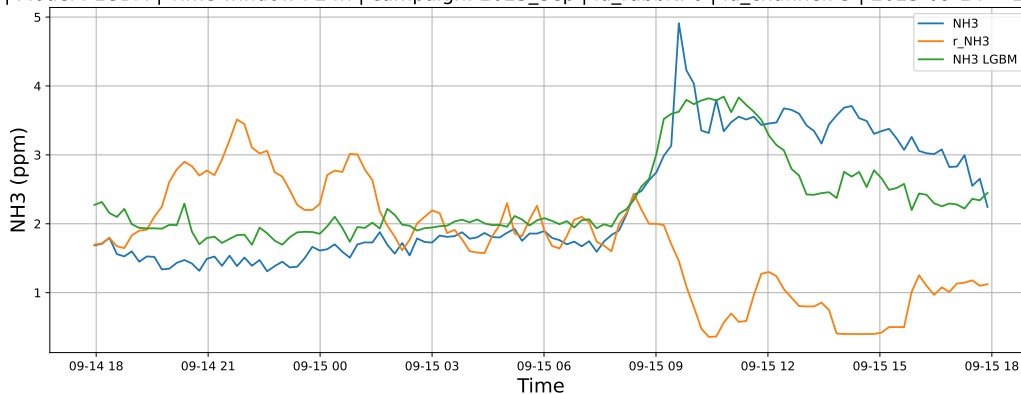
4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

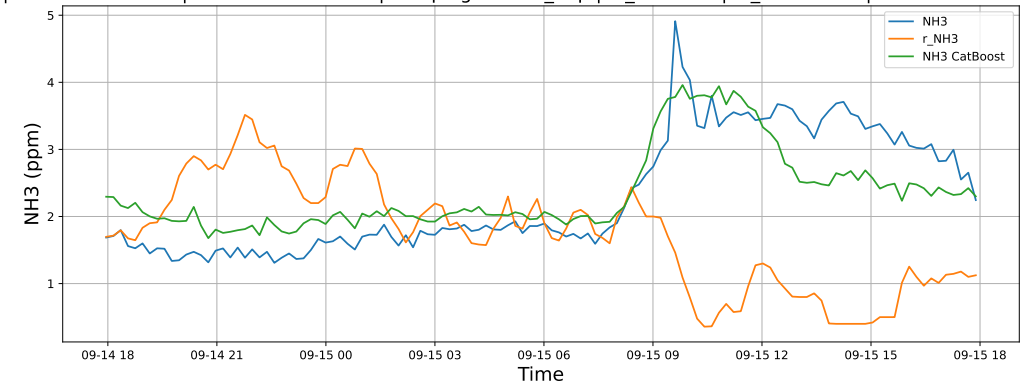
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

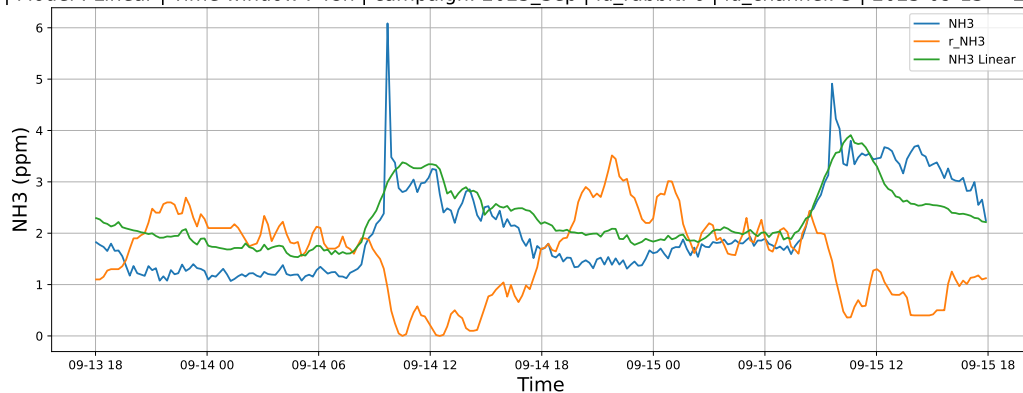


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

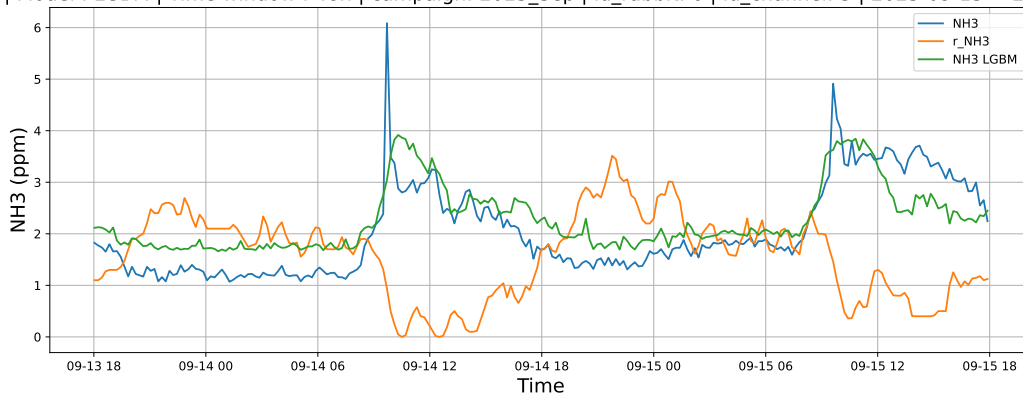


4.1.8.2 Time window : 48

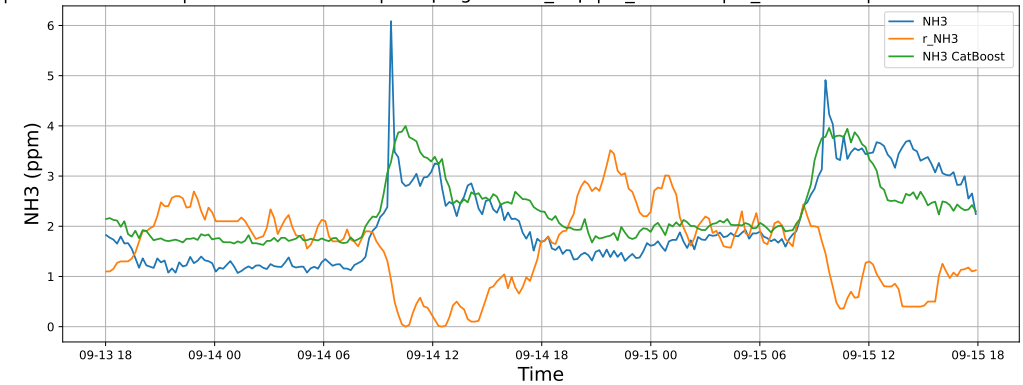
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

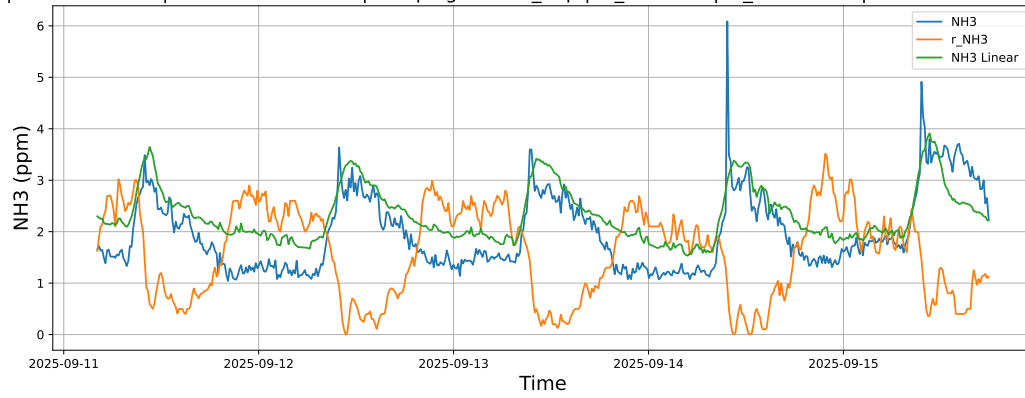


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

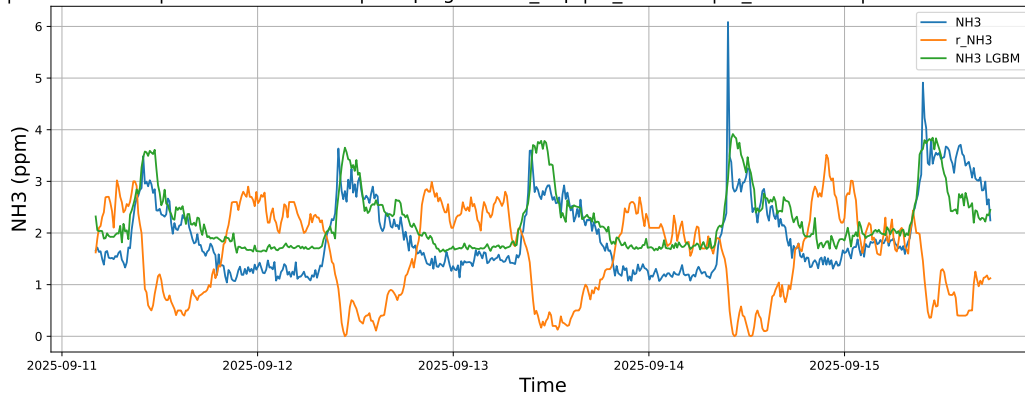


4.1.8.3 Time window : ALL

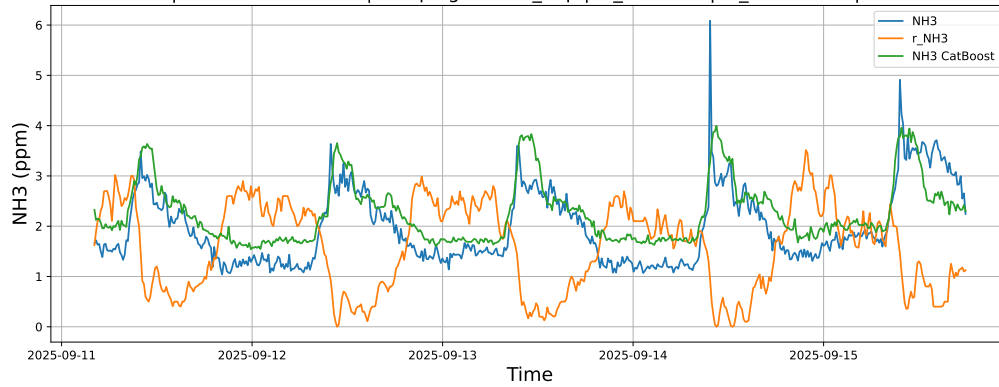
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

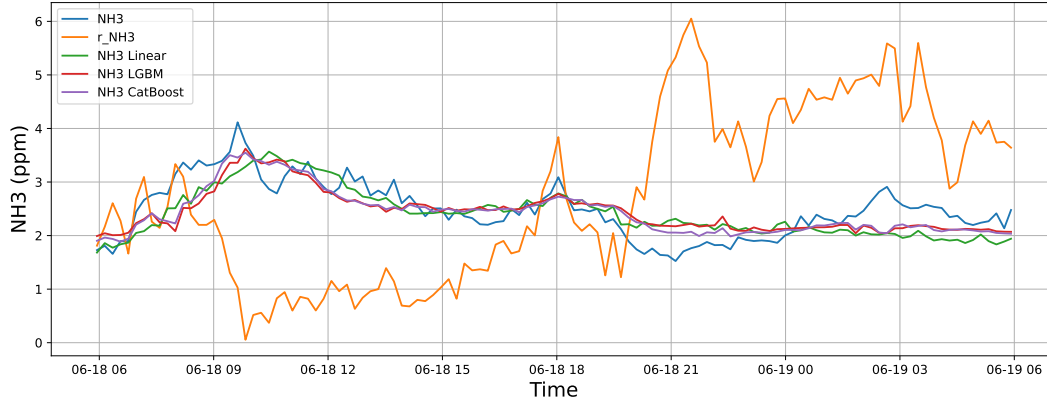


4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

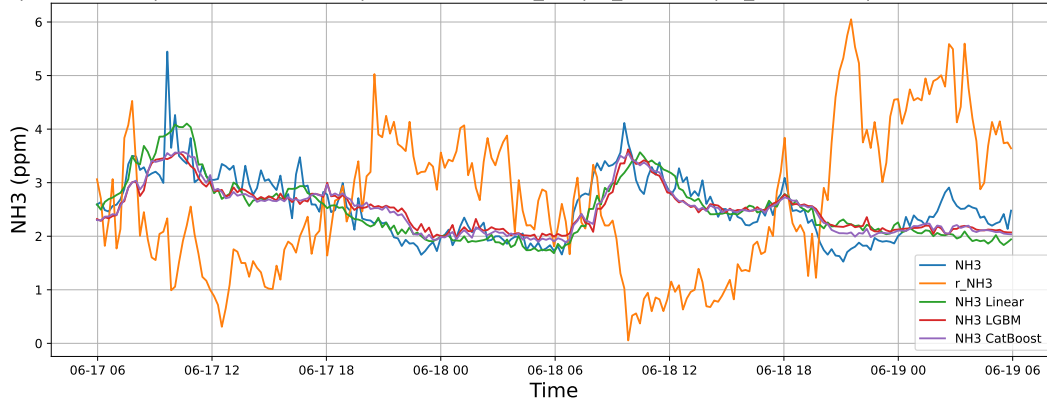
4.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



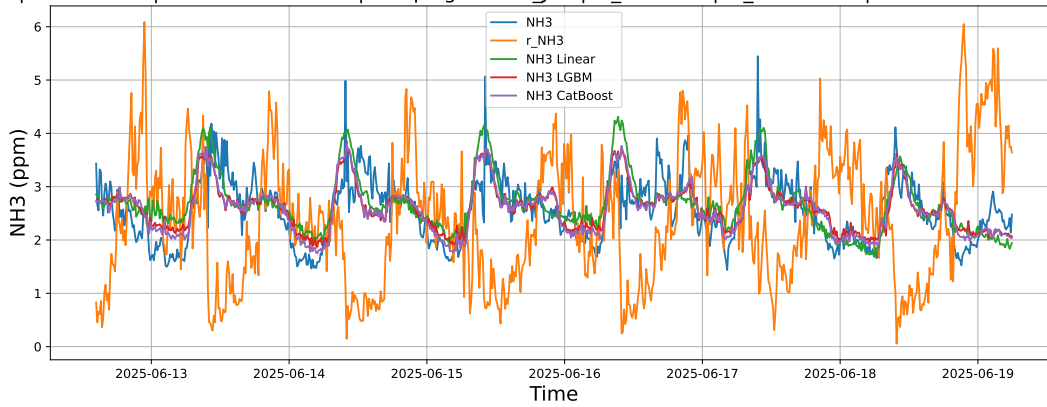
4.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



4.2.1.3 Time window : ALL

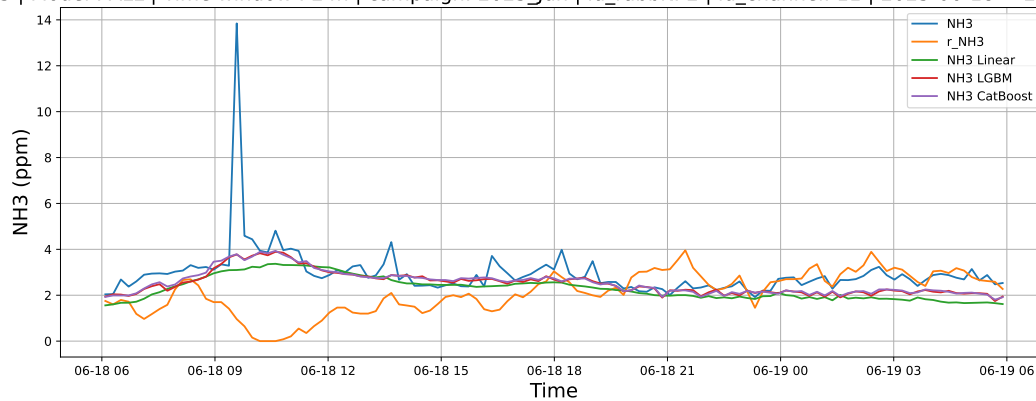
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

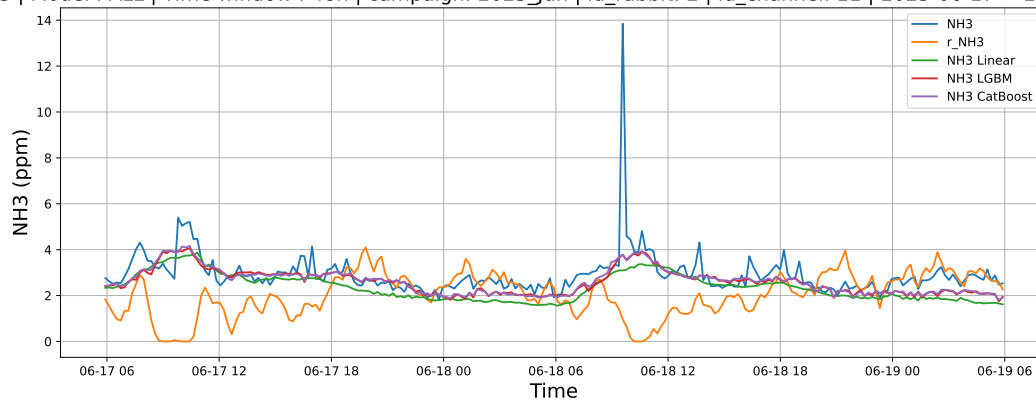
4.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



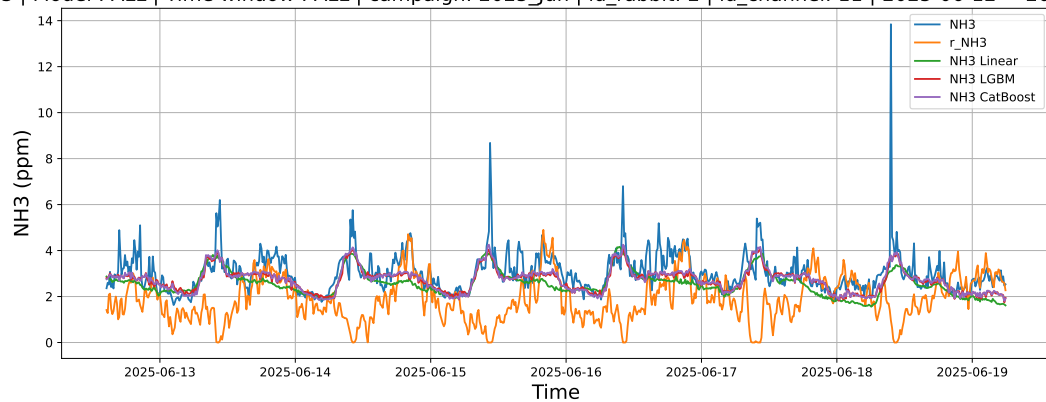
4.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



4.2.2.3 Time window : ALL

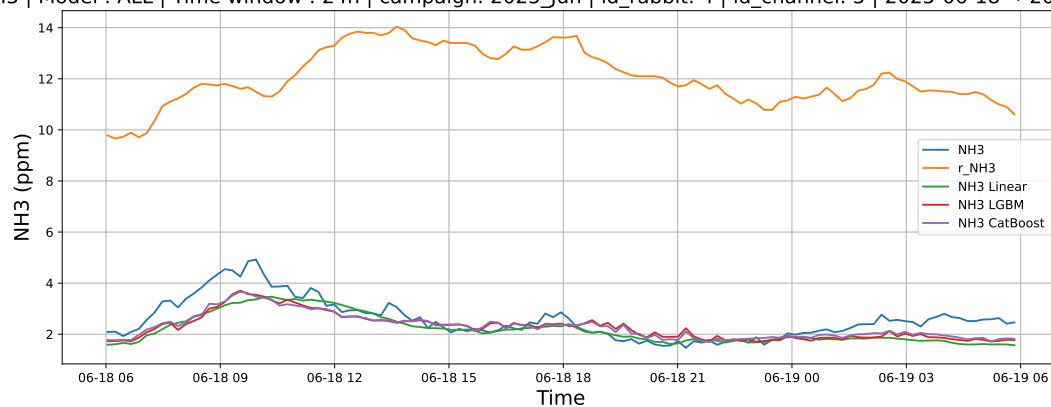
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

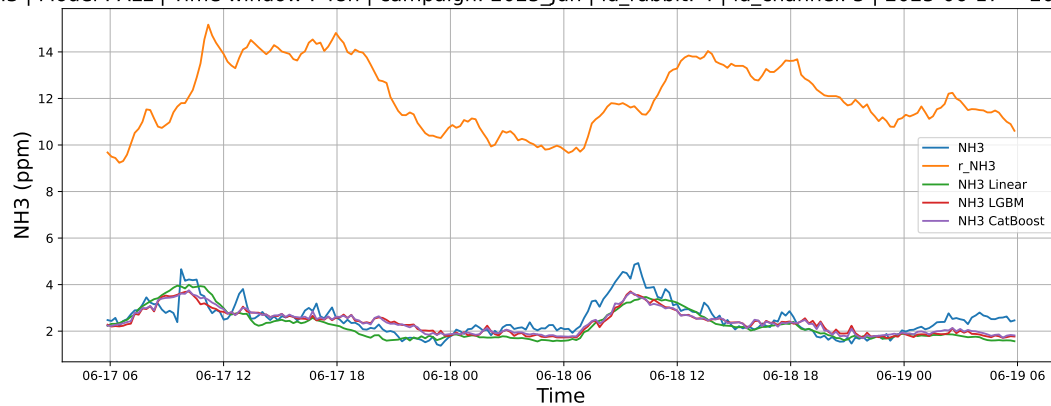
4.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



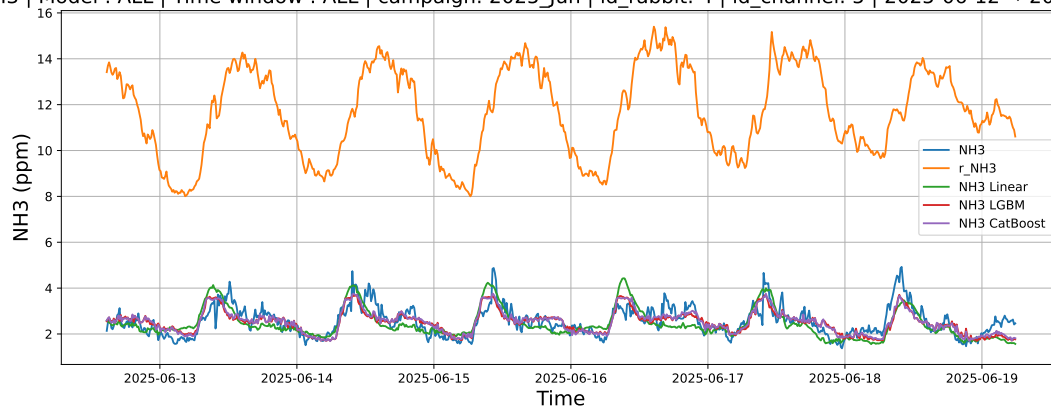
4.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



4.2.3.3 Time window : ALL

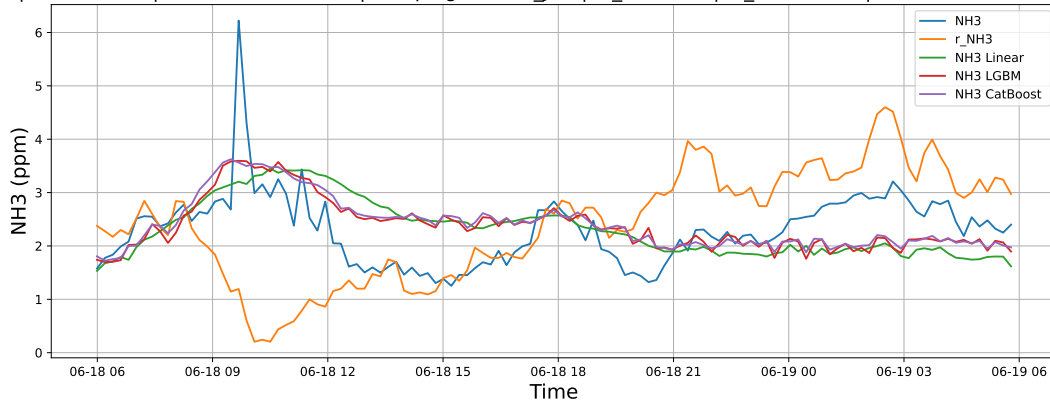
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

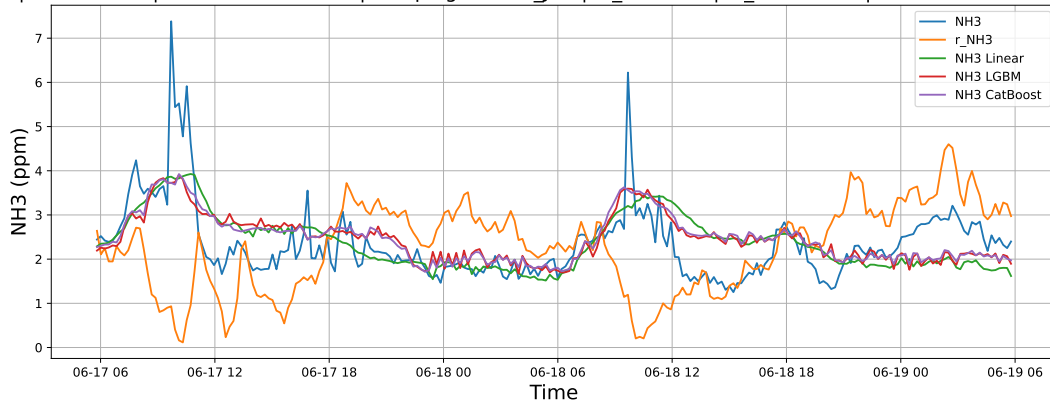
4.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



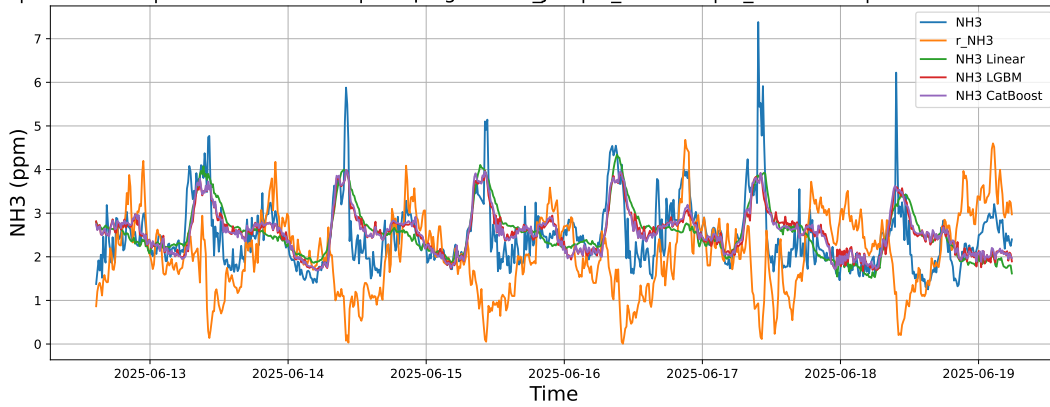
4.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



4.2.4.3 Time window : ALL

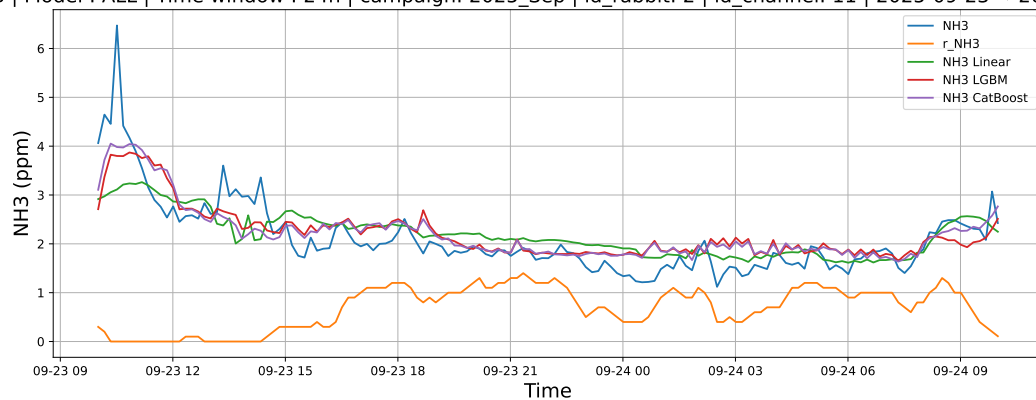
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

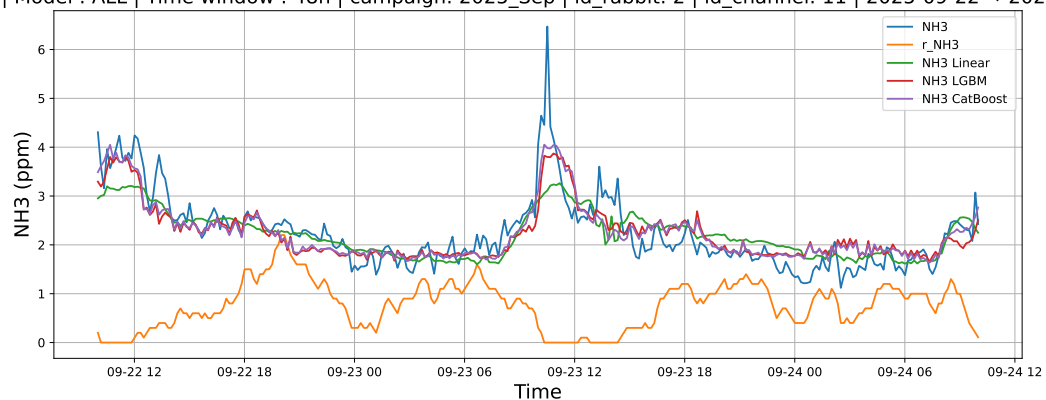
4.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



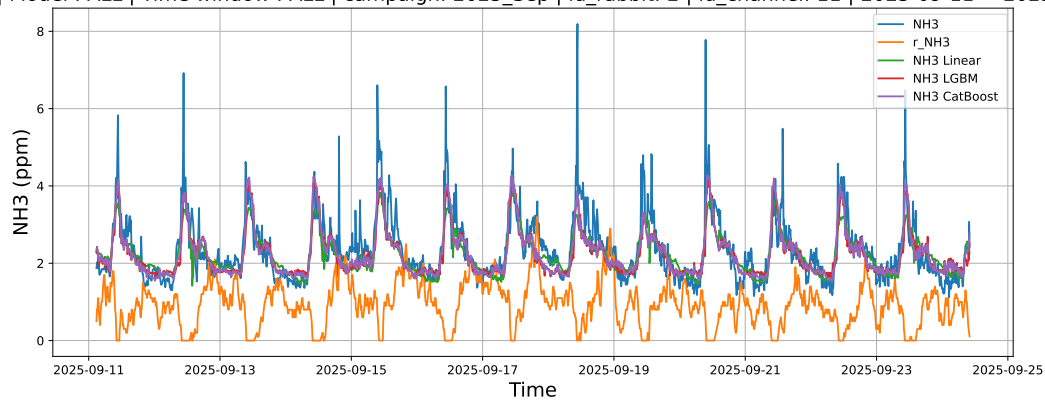
4.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



4.2.5.3 Time window : ALL

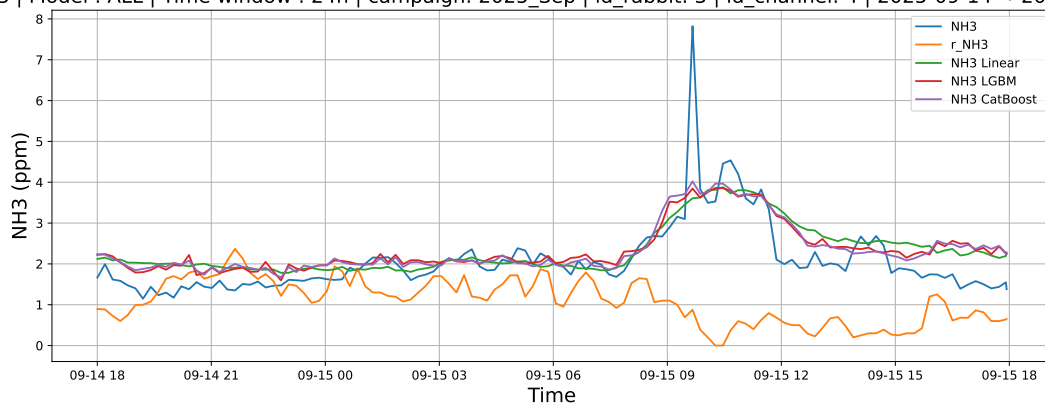
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

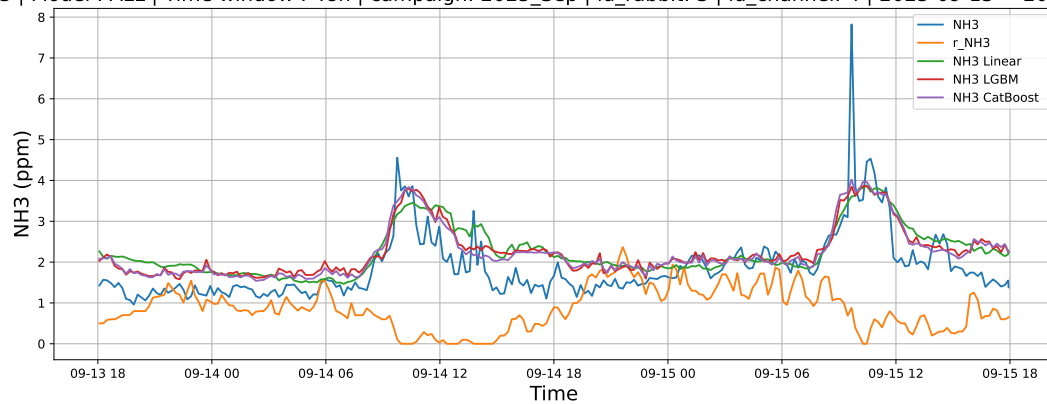
4.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



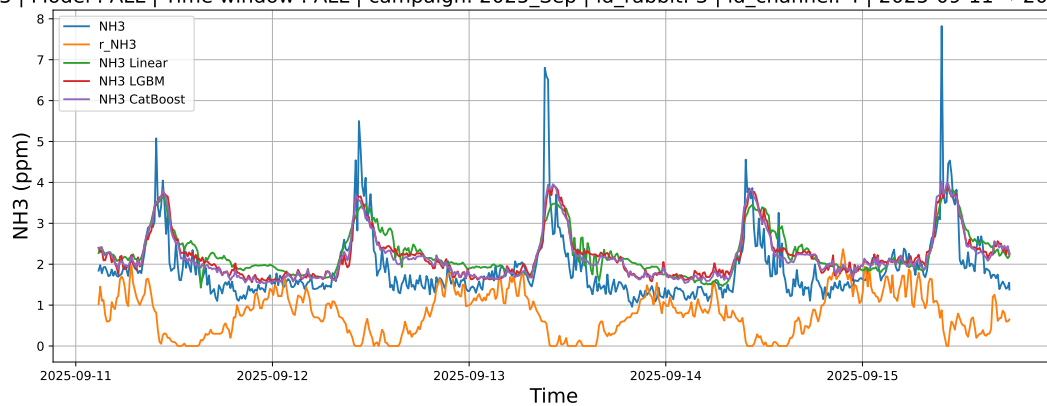
4.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



4.2.6.3 Time window : ALL

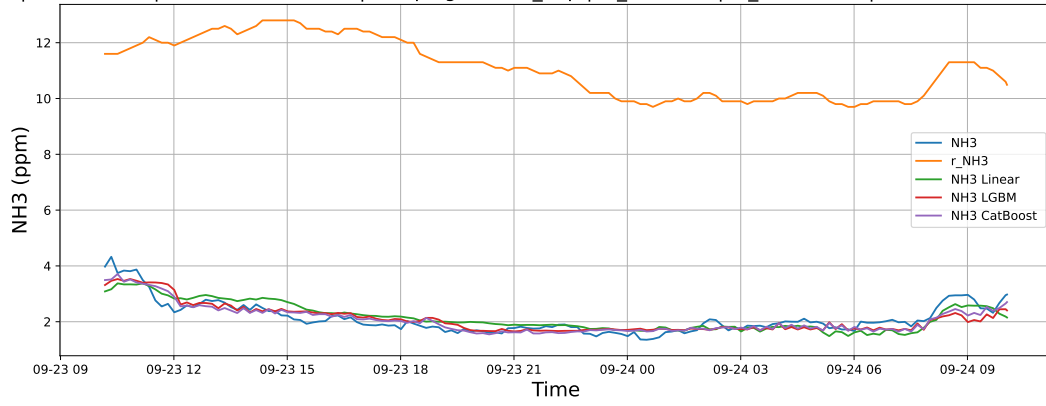
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

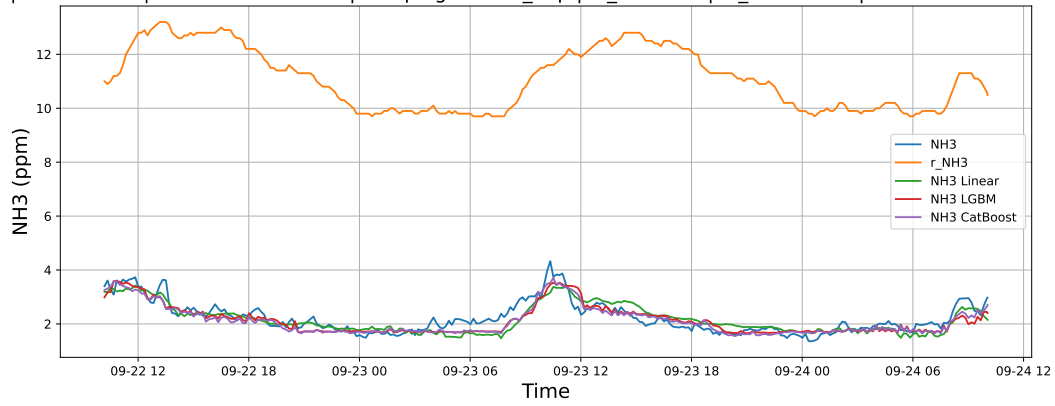
4.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



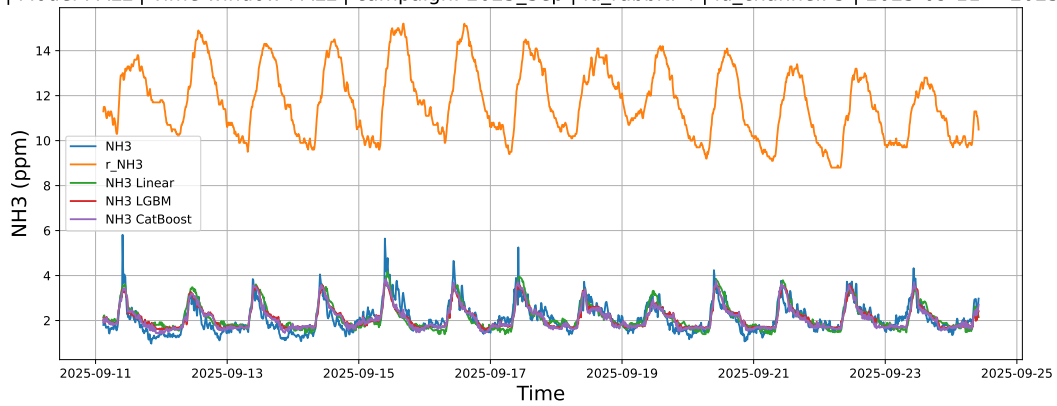
4.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



4.2.7.3 Time window : ALL

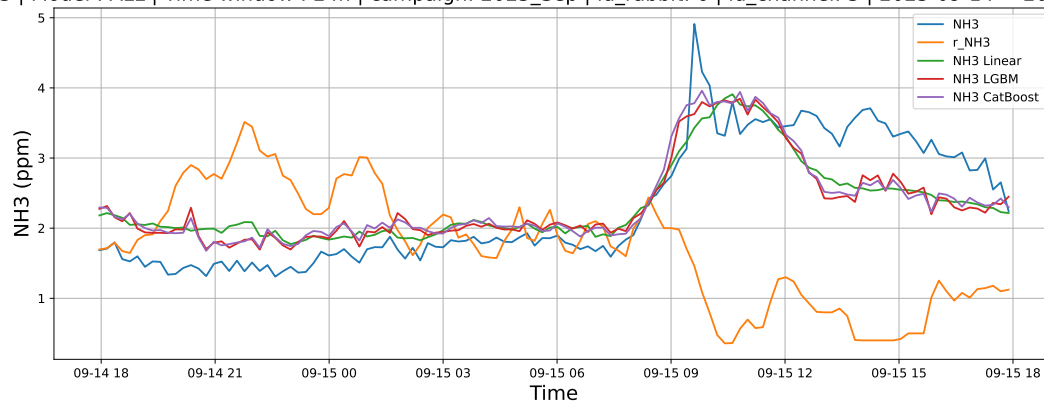
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

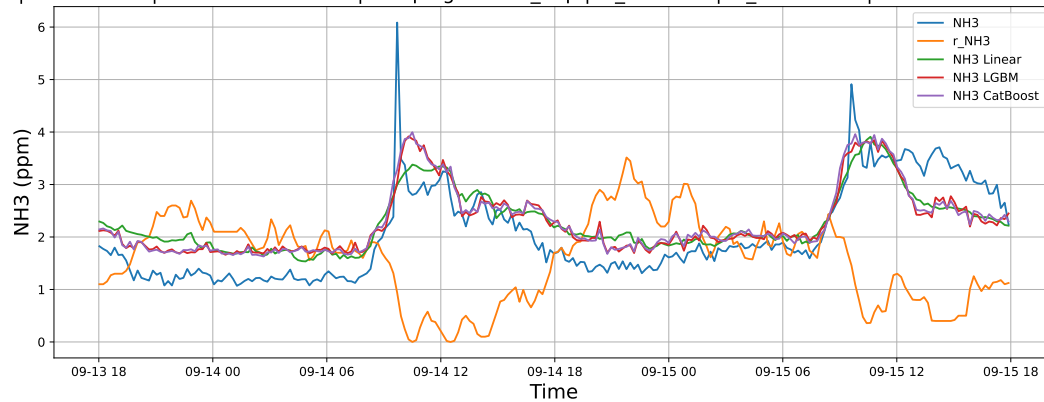
4.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



4.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



4.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

